

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

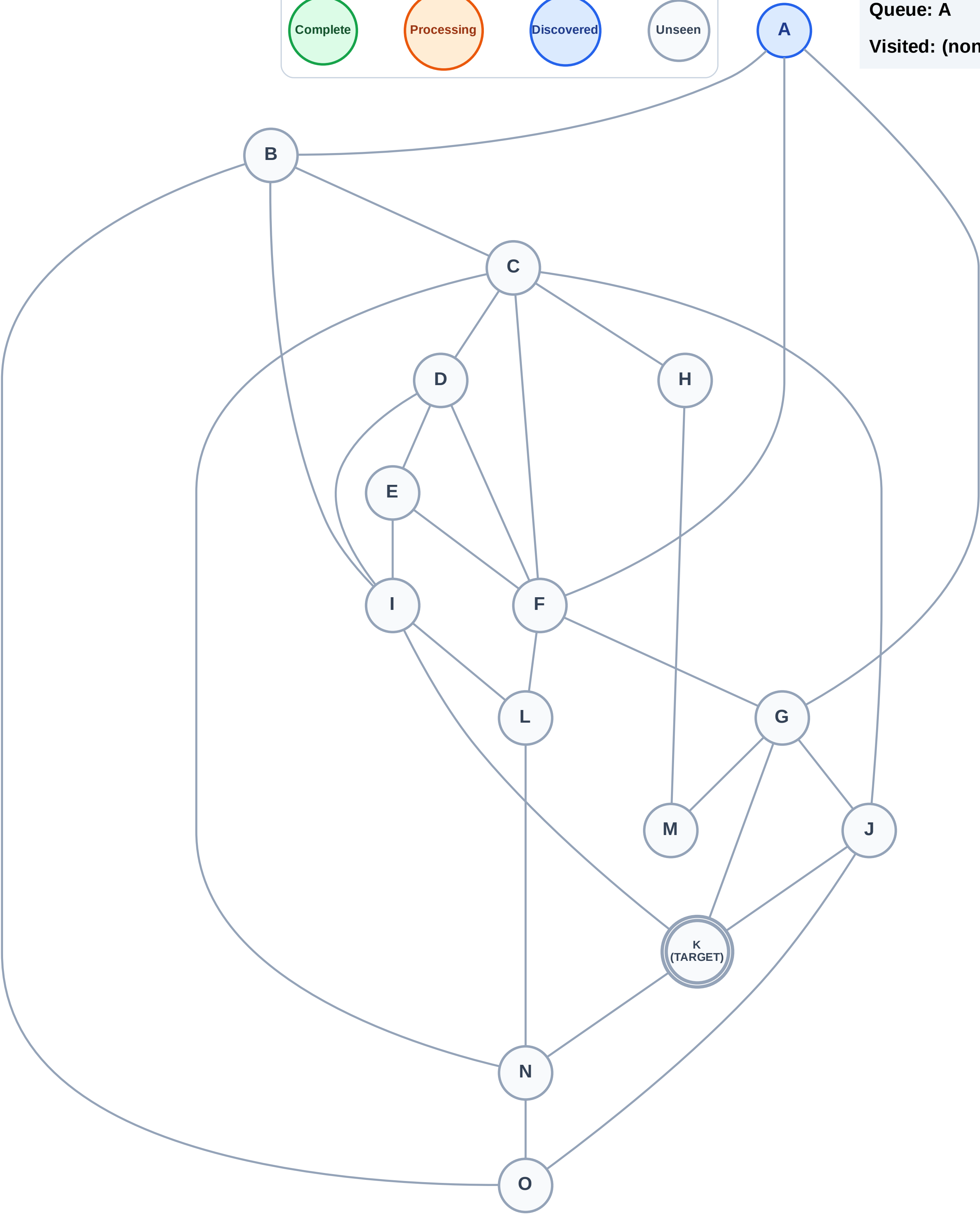
Complete

Processing

Discovered

Unseen

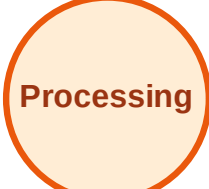
Queue: A
Visited: (none)



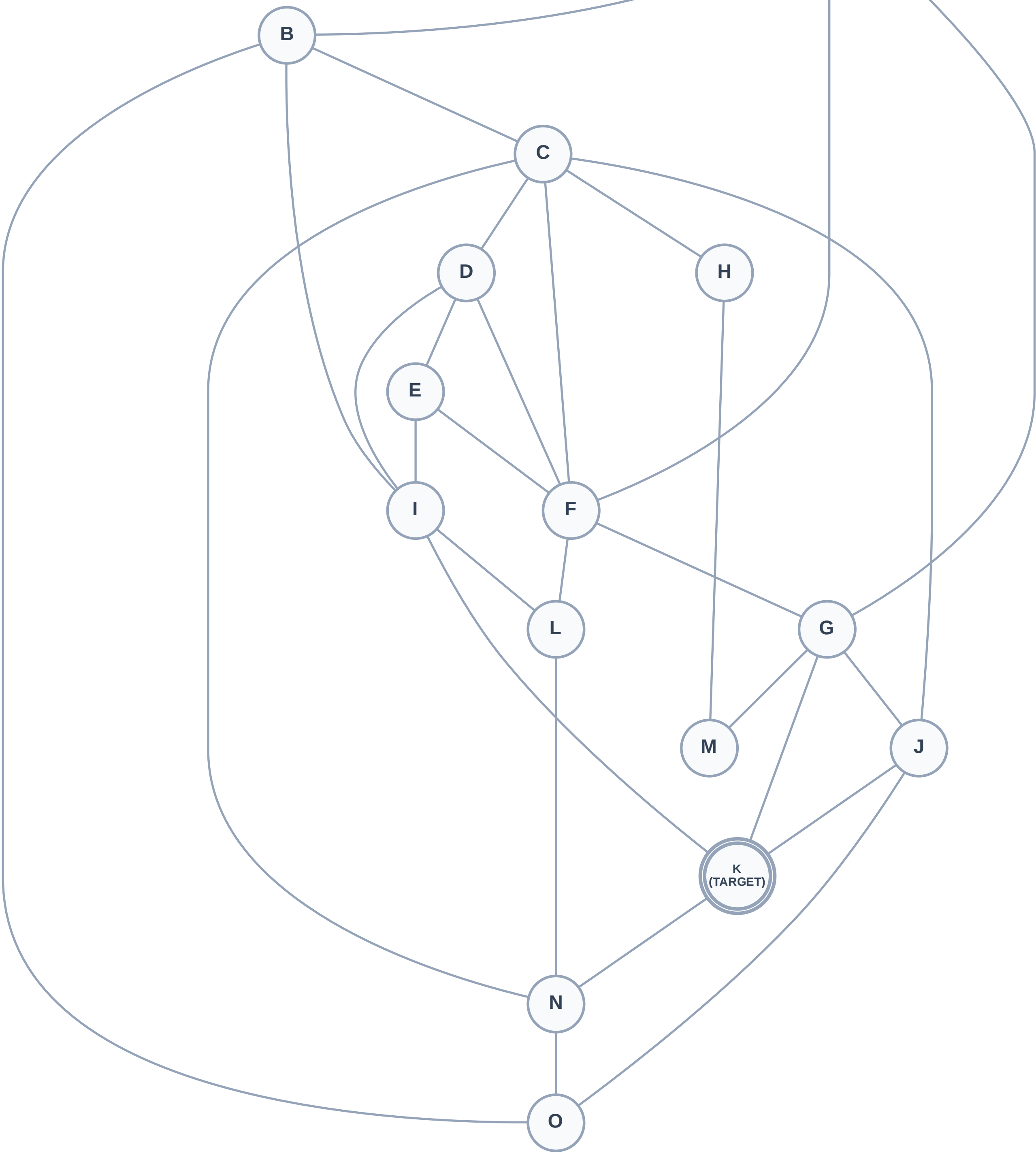
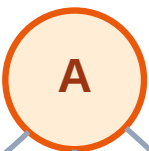
BFS Traversal

Step 2: Process node A

Legend



Queue: (empty)
Visited: (none)



BFS Traversal

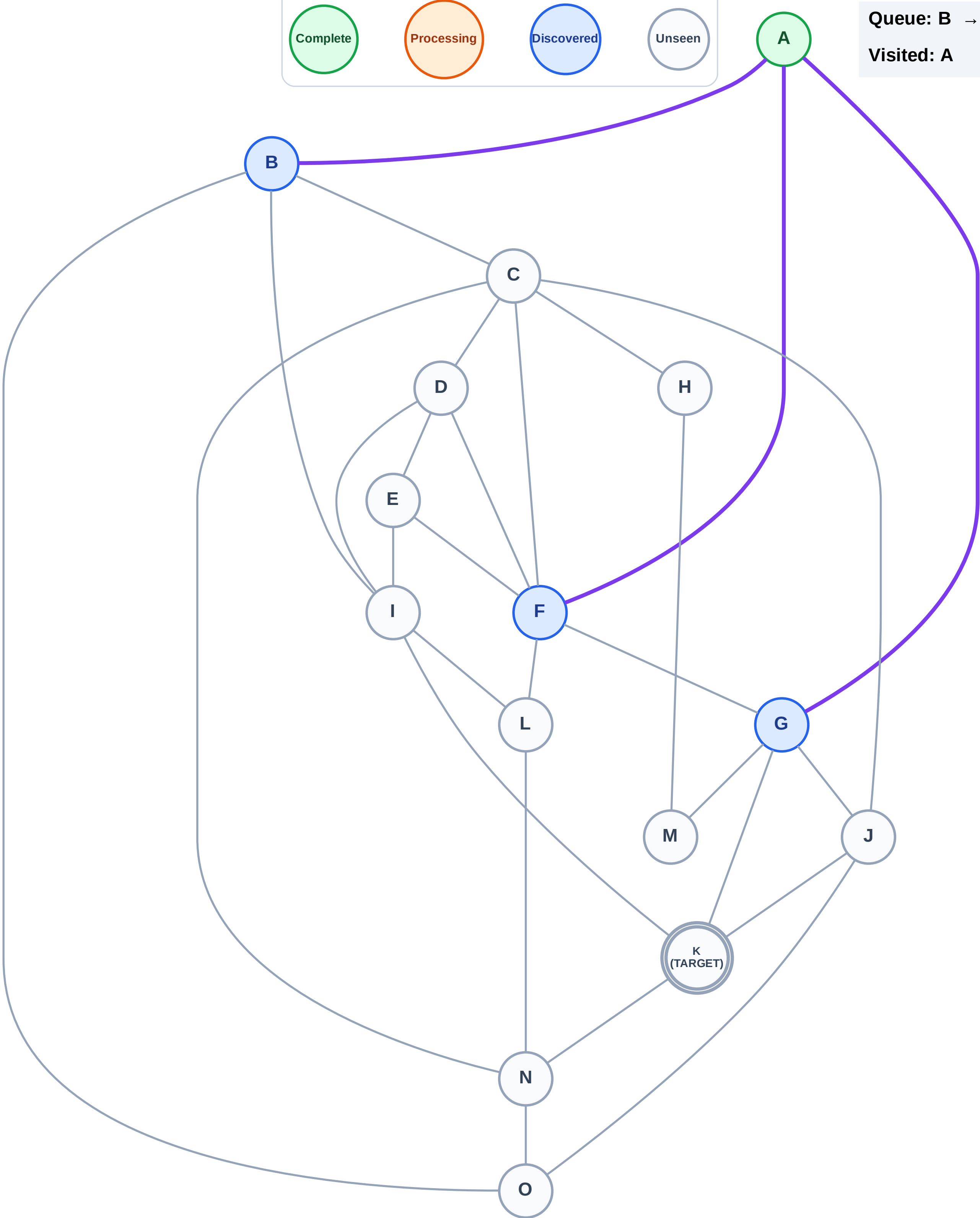
Step 3: Discover B, F, G from A

Legend



Queue: B → F → G

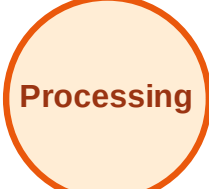
Visited: A



BFS Traversal

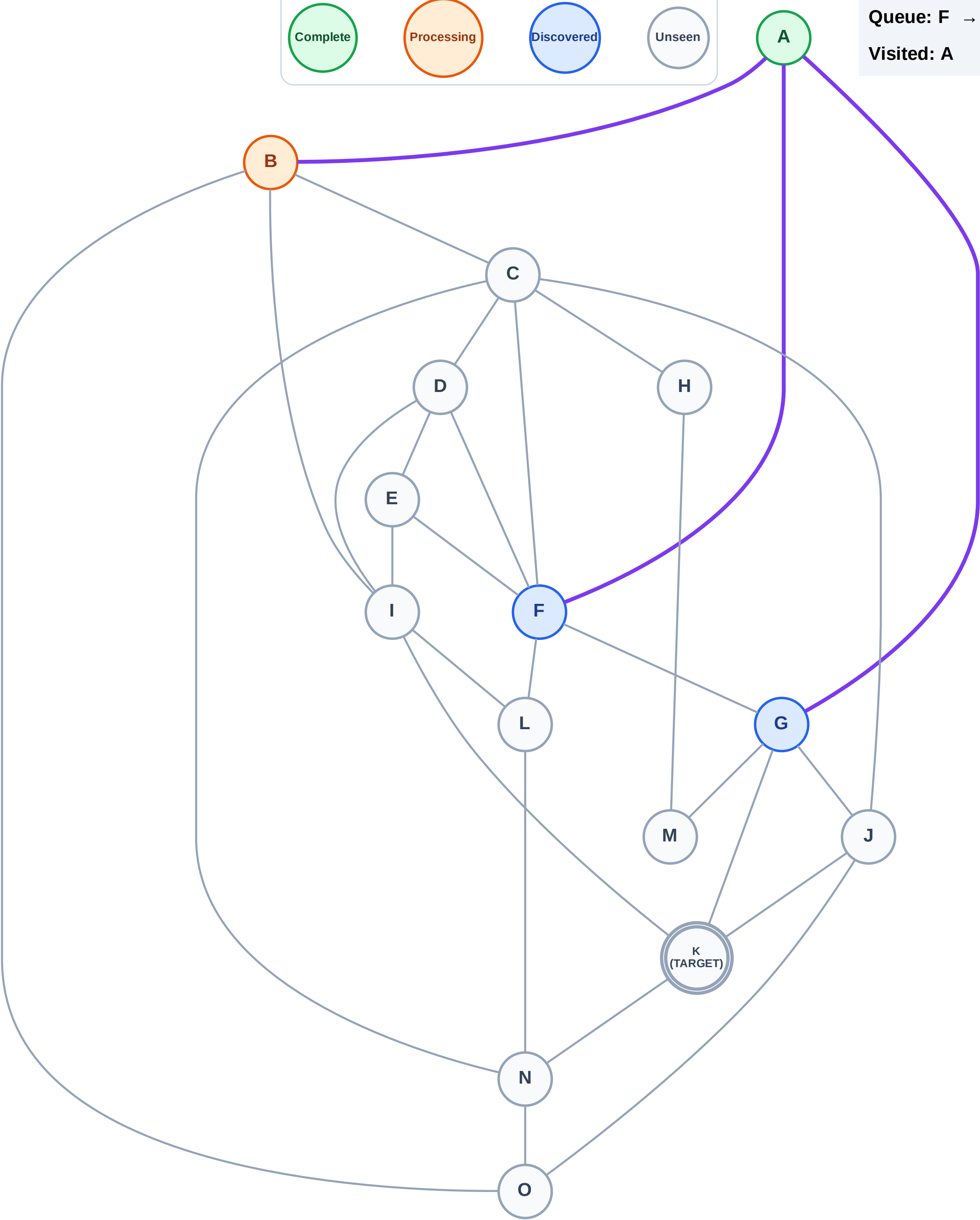
Step 4: Process node B

Legend



Queue: F → G

Visited: A



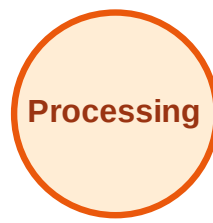
BFS Traversal

Step 5: Discover C, I, O from B

Legend



Complete



Processing

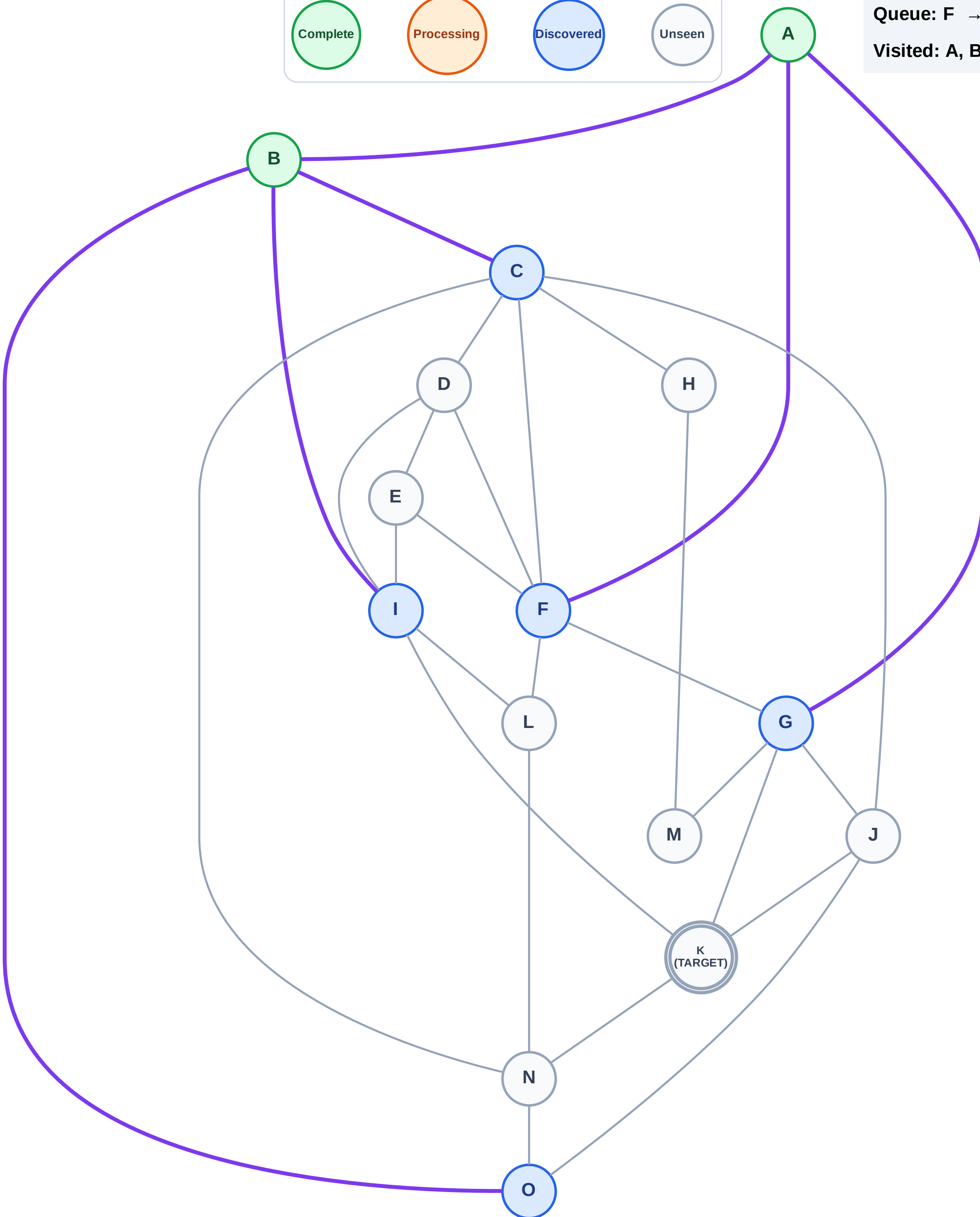


Discovered



Unseen

Queue: F → G → C → I → O
Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

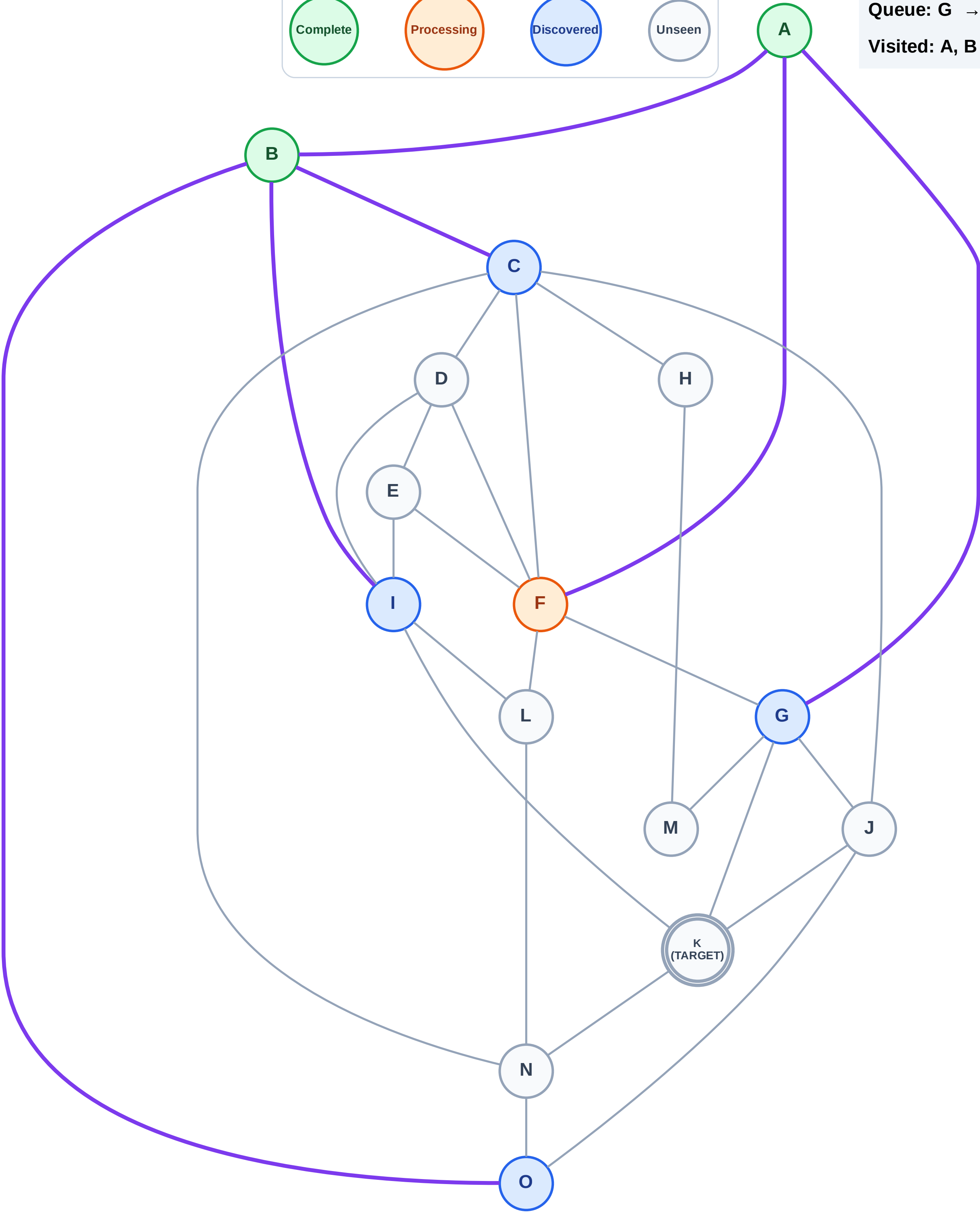
Processing

Discovered

Unseen

Queue: G → C → I → O

Visited: A, B



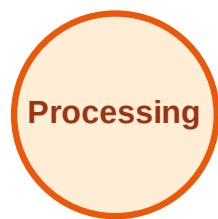
BFS Traversal

Step 7: Discover D, E, L from F

Legend



Complete



Processing

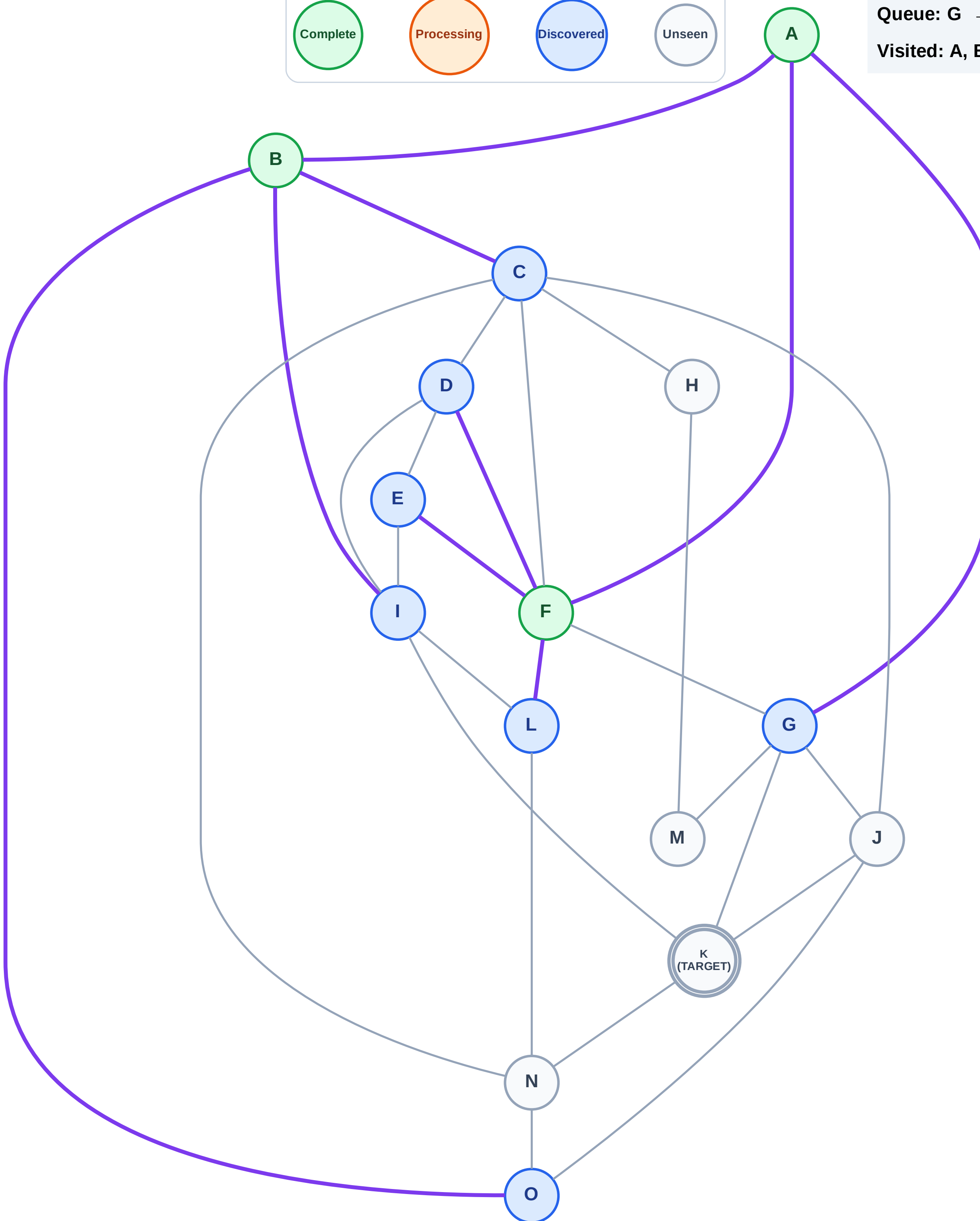


Discovered



Unseen

Queue: G → C → I → O → D → E → L
Visited: A, B, F



BFS Traversal

Step 8: Process node G

Legend

Complete

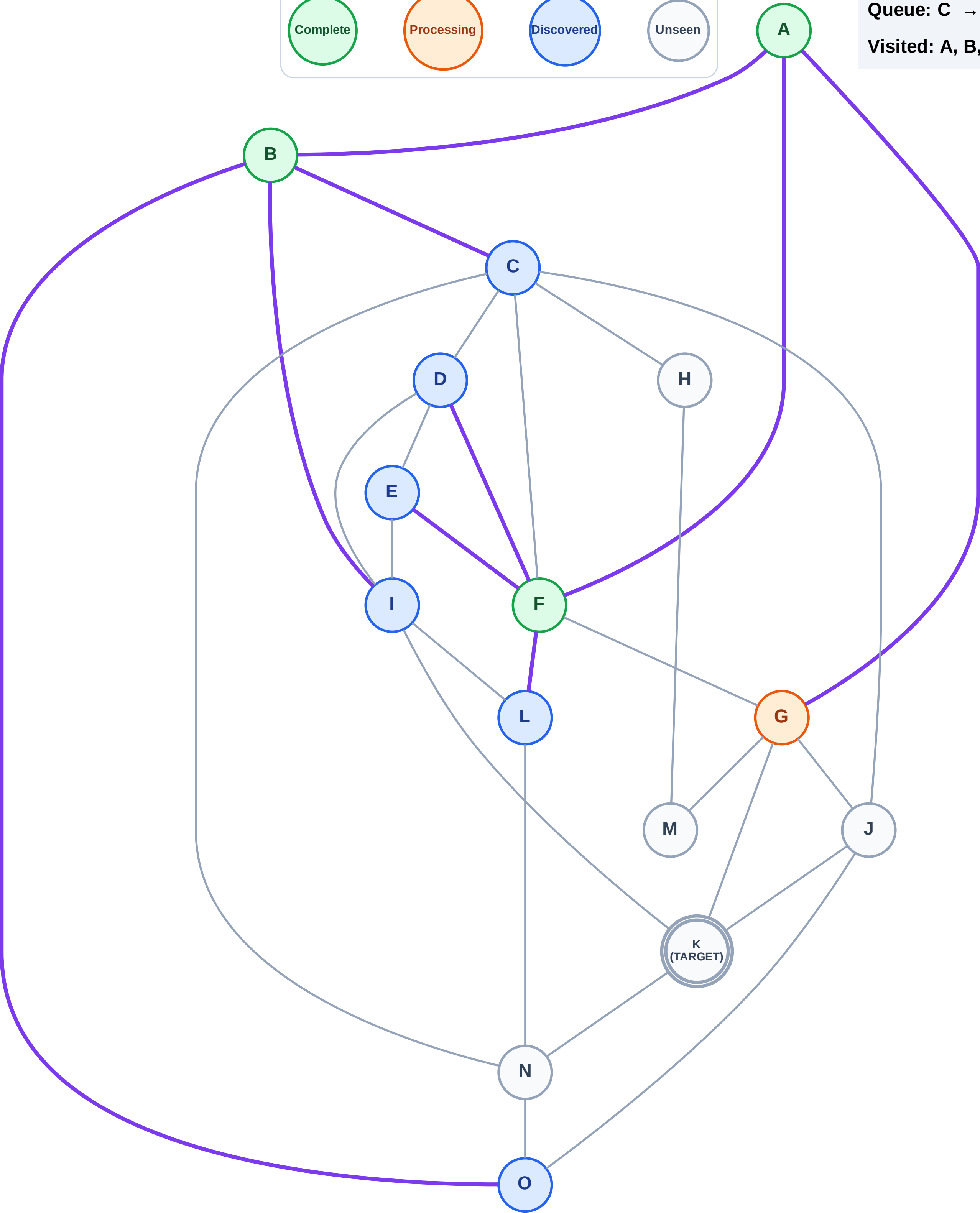
Processing

Discovered

Unseen

Queue: C → I → O → D → E → L

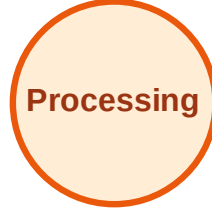
Visited: A, B, F



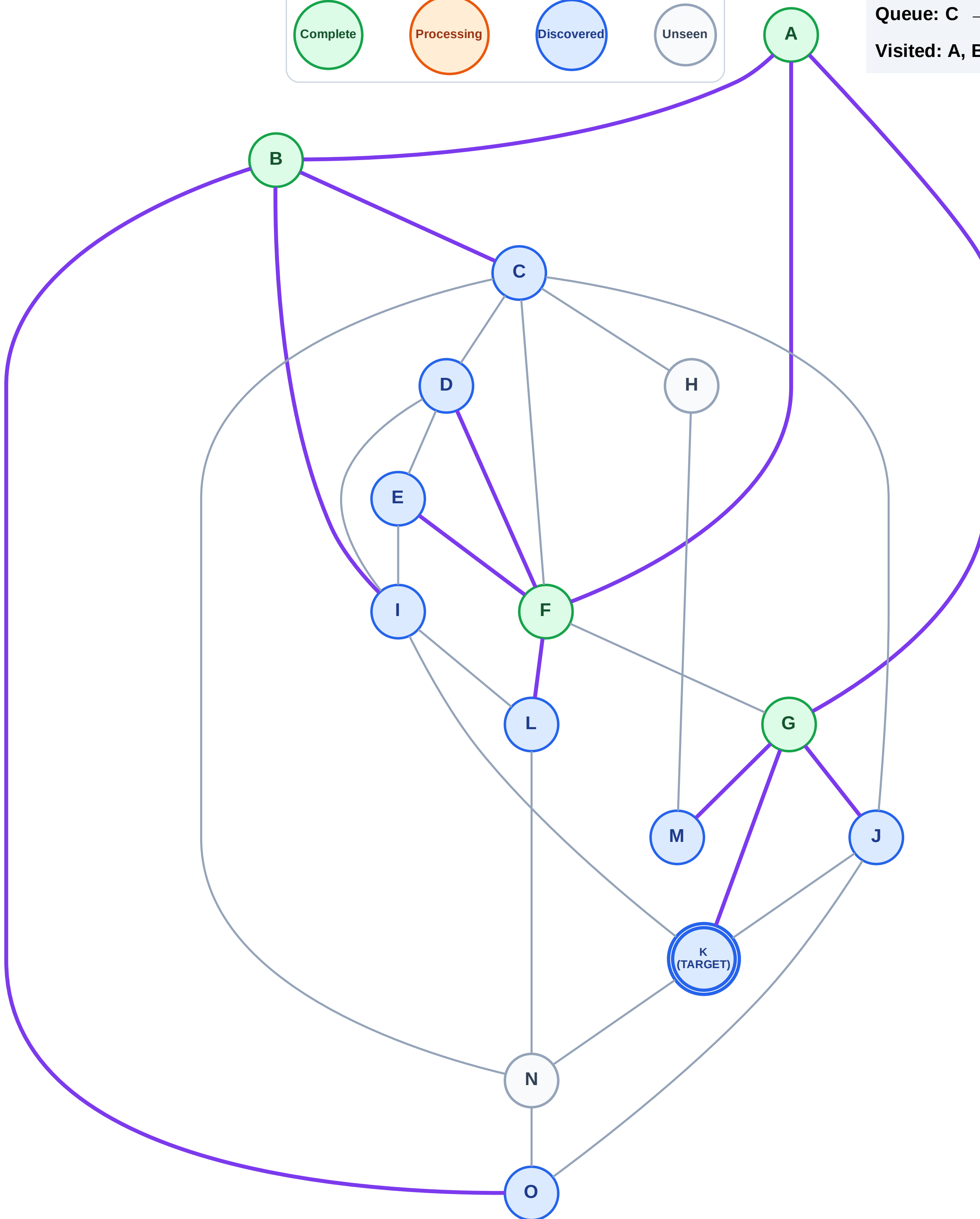
BFS Traversal

Step 9: Discover J, K, M from G

Legend



Queue: C → I → O → D → E → L → J → K → M
Visited: A, B, F, G



BFS Traversal

Step 10: Process node C

Legend

Complete

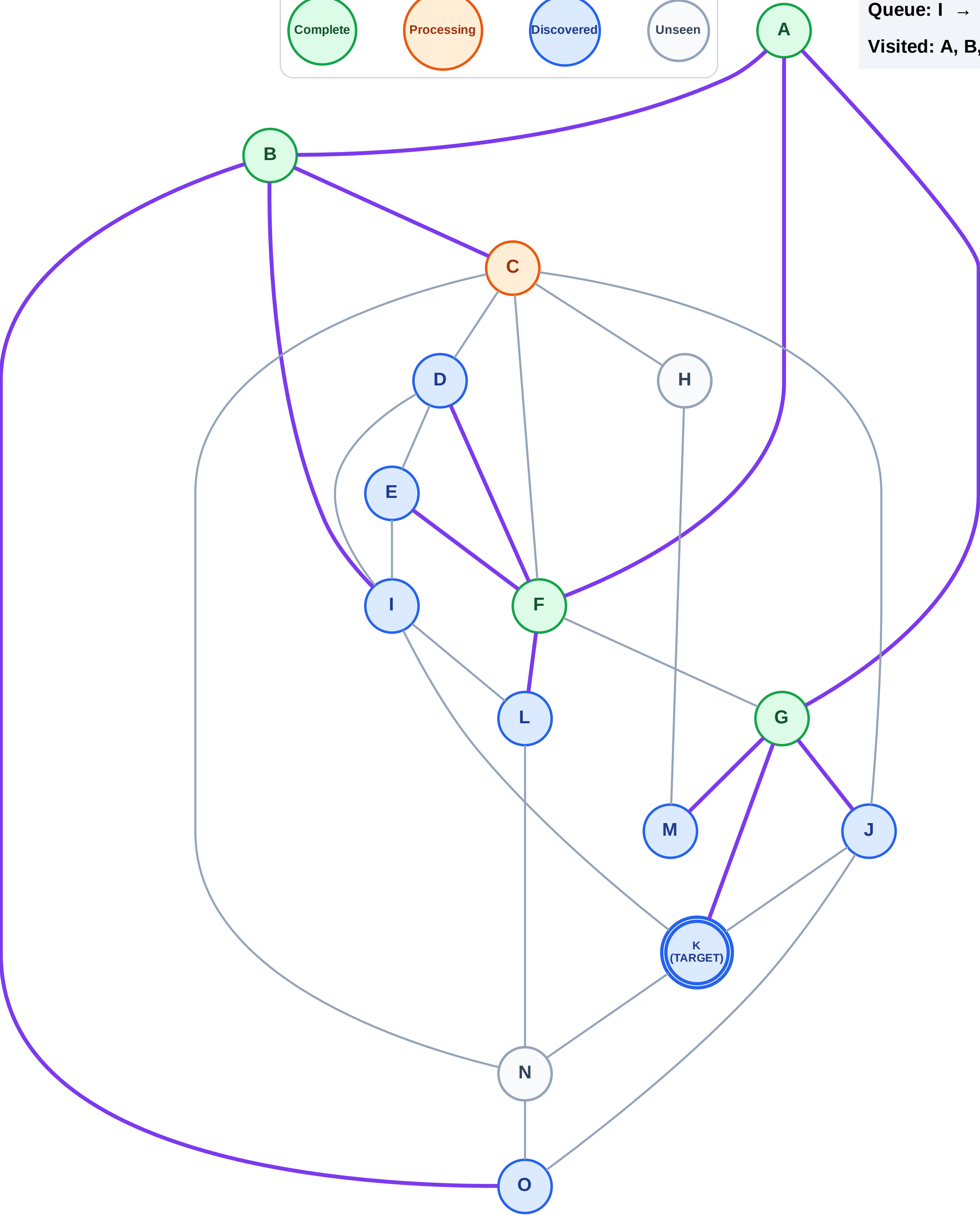
Processing

Discovered

Unseen

Queue: I → O → D → E → L → J → K → M

Visited: A, B, F, G



BFS Traversal

Step 11: Discover H, N from C

Legend



Complete



Processing



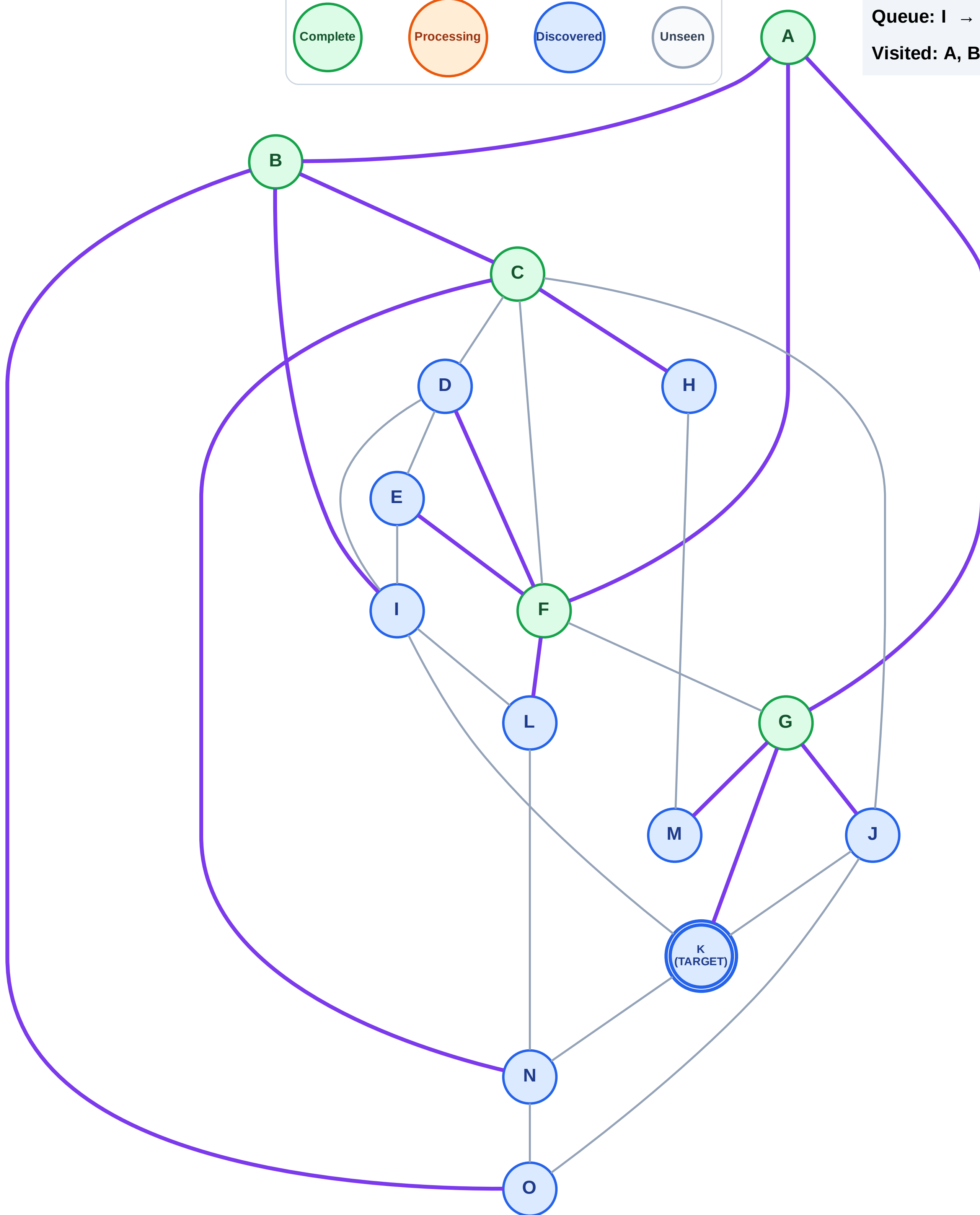
Discovered



Unseen

Queue: I → O → D → E → L → J → K → M → H → N

Visited: A, B, C, F, G



BFS Traversal

Step 12: Process node I

Legend

Complete

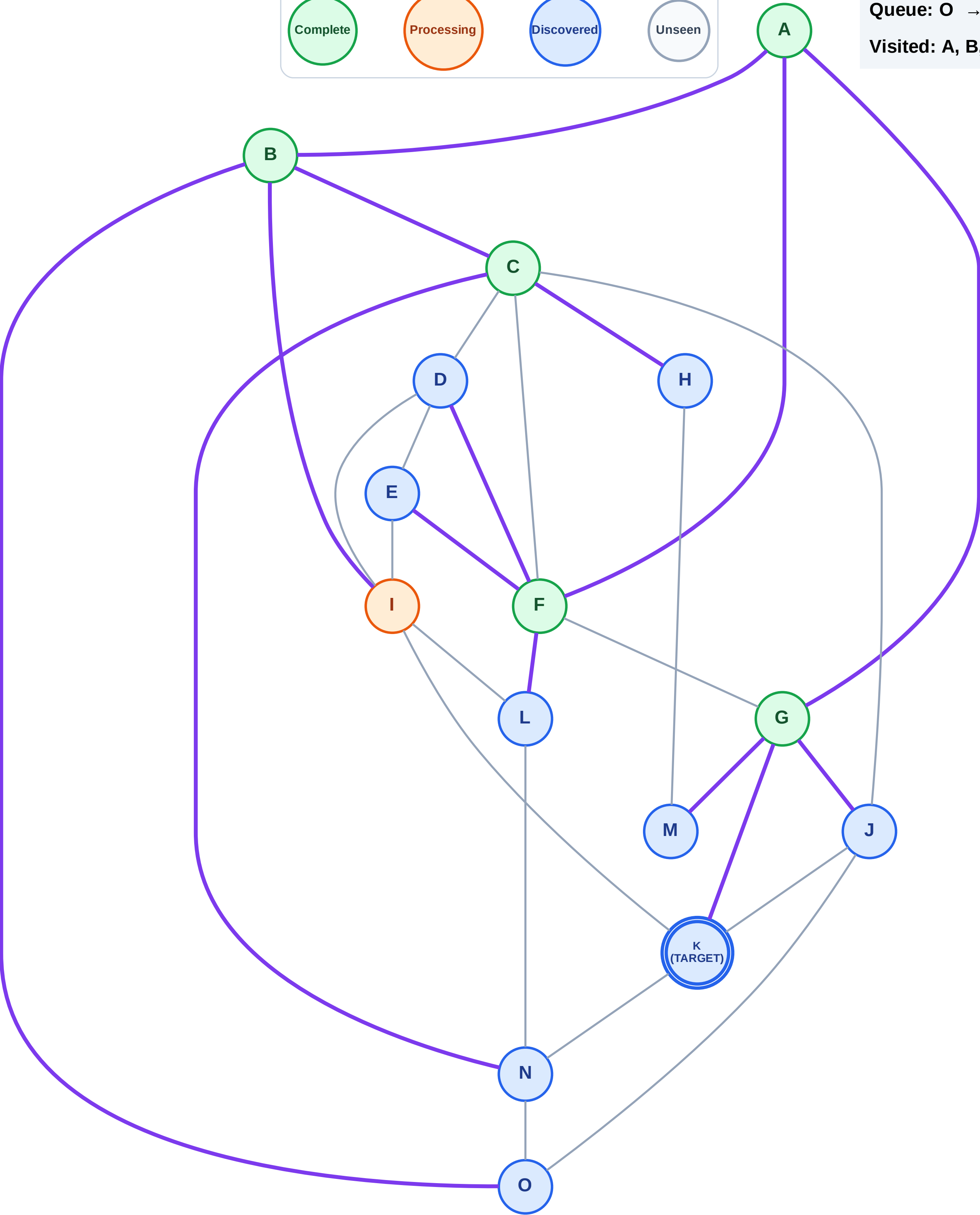
Processing

Discovered

Unseen

Queue: O → D → E → L → J → K → M → H → N

Visited: A, B, C, F, G



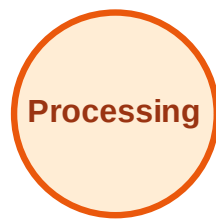
BFS Traversal

Step 13: No new neighbors from I

Legend



Complete



Processing

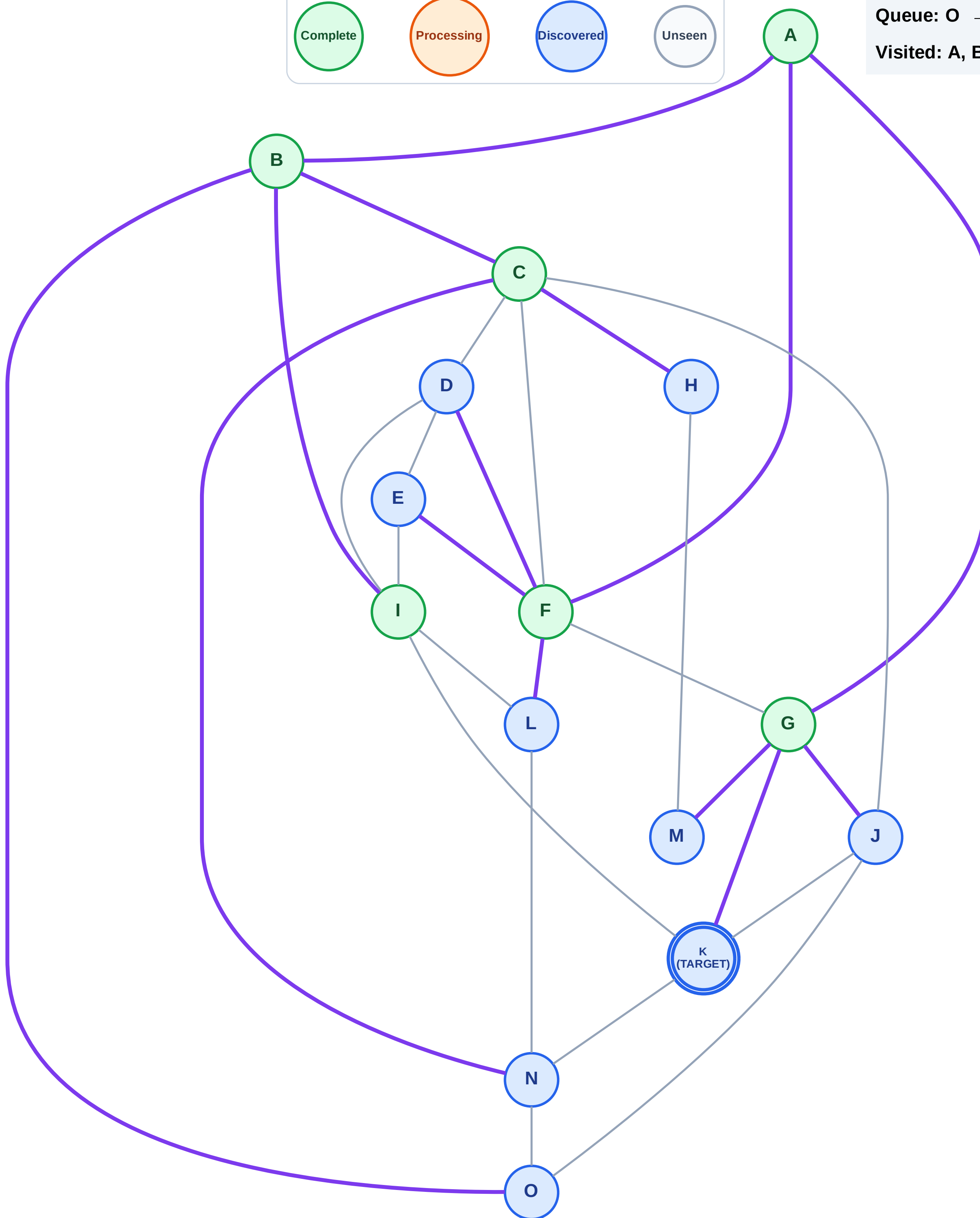


Discovered



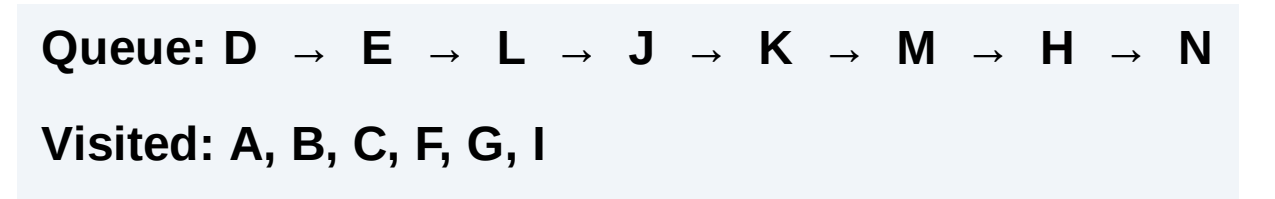
Unseen

Queue: O → D → E → L → J → K → M → H → N
Visited: A, B, C, F, G, I



Step 14: Process node O

Page 10 of 10



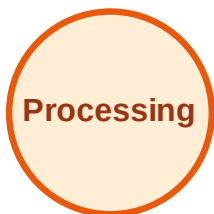
BFS Traversal

Step 15: No new neighbors from O

Legend



Complete



Processing



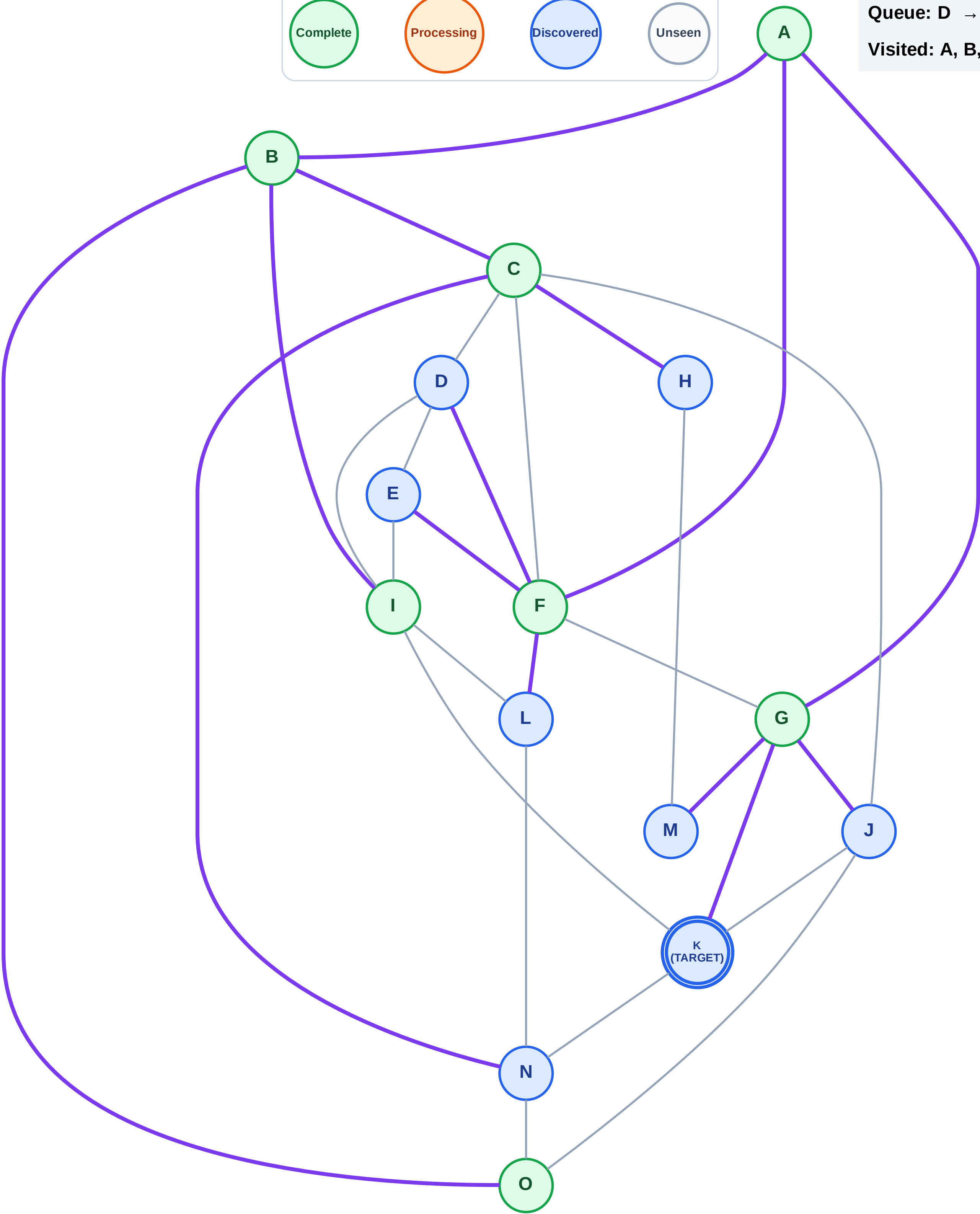
Discovered



Unseen

Queue: D → E → L → J → K → M → H → N

Visited: A, B, C, F, G, I, O



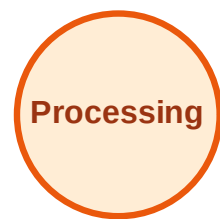
BFS Traversal

Step 16: Process node D

Legend



Complete



Processing



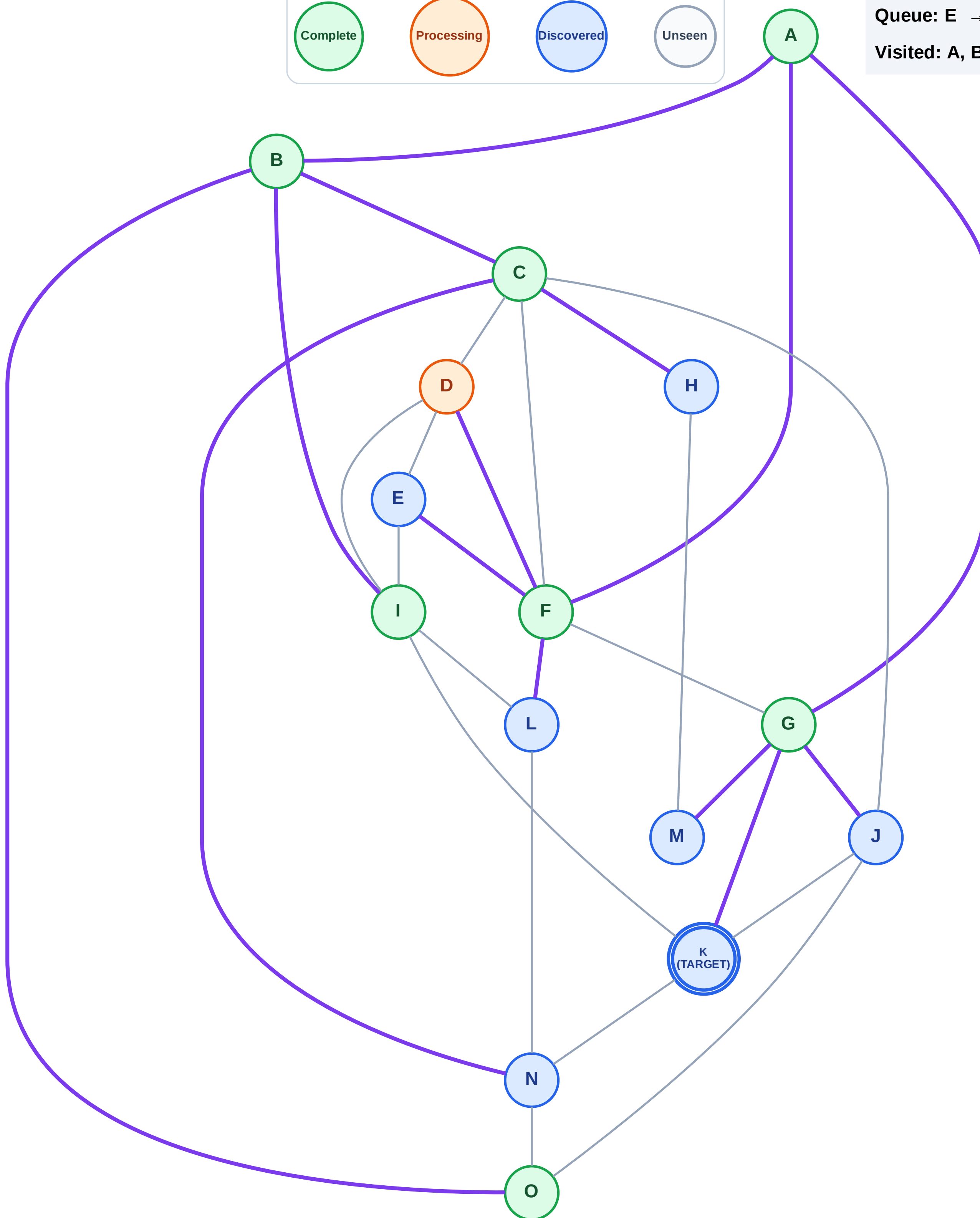
Discovered



Unseen

Queue: E → L → J → K → M → H → N

Visited: A, B, C, F, G, I, O



BFS Traversal

Step 17: No new neighbors from D

Legend

Complete

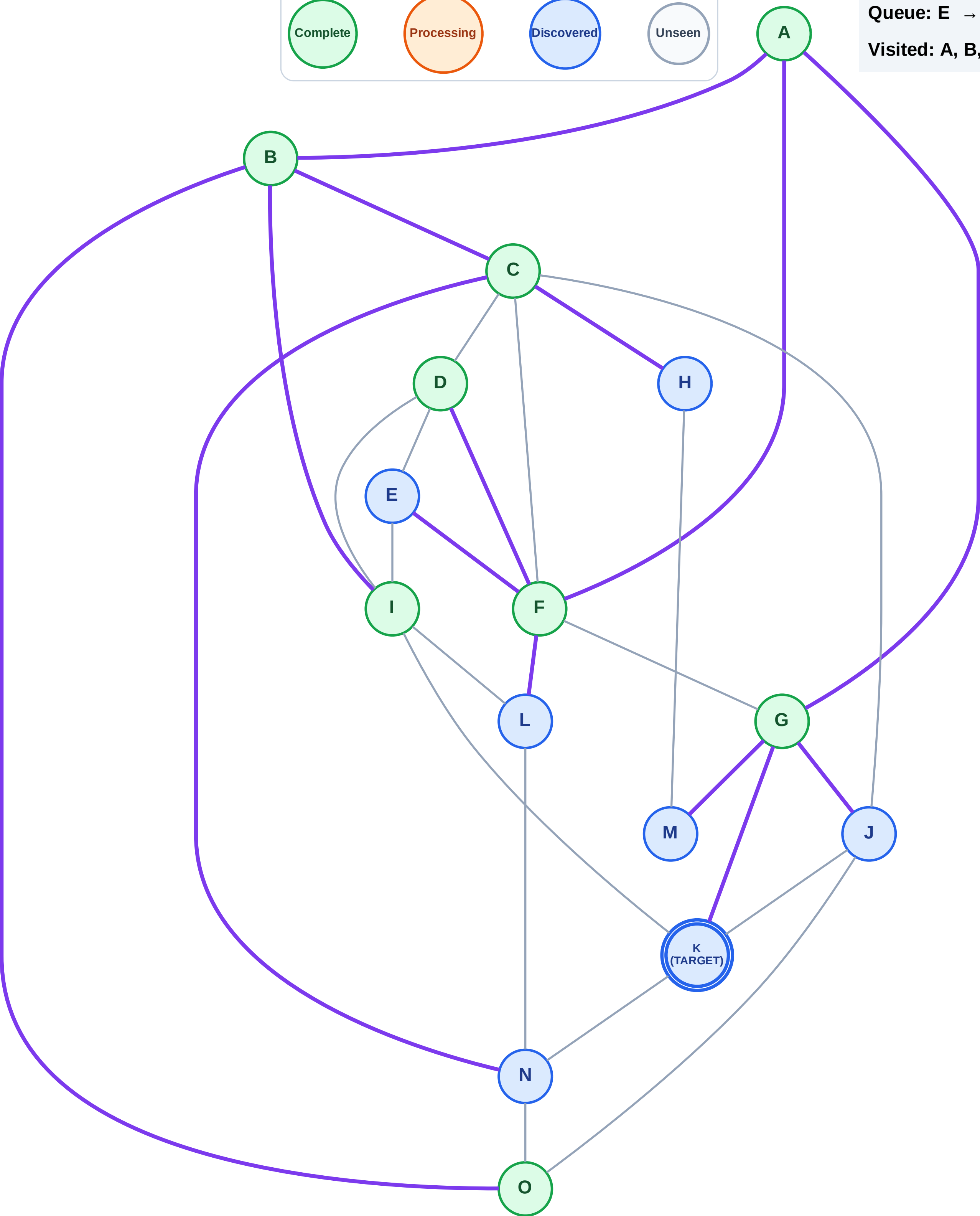
Processing

Discovered

Unseen

Queue: E → L → J → K → M → H → N

Visited: A, B, C, D, F, G, I, O



BFS Traversal

Step 18: Process node E

Legend

Complete

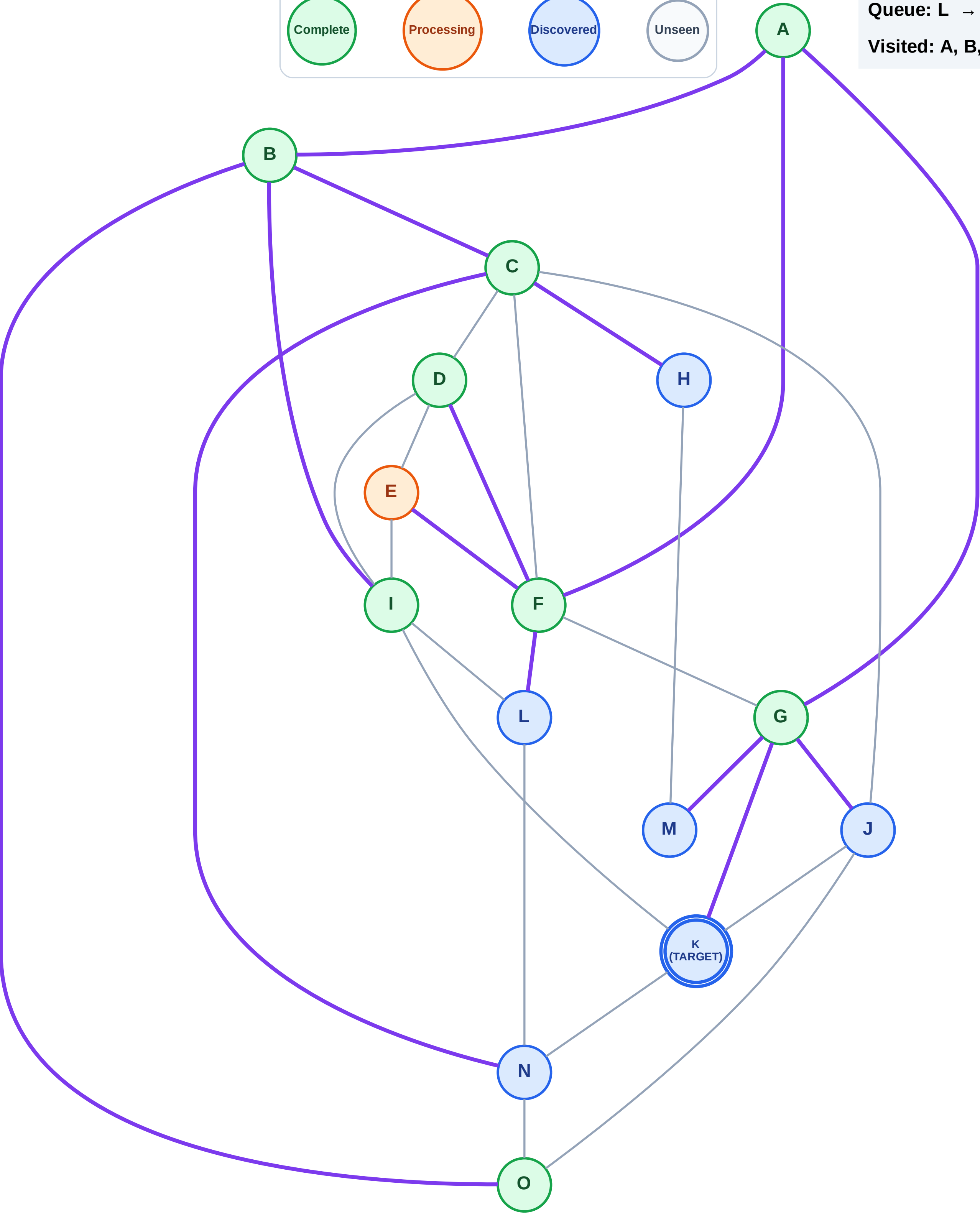
Processing

Discovered

Unseen

Queue: L → J → K → M → H → N

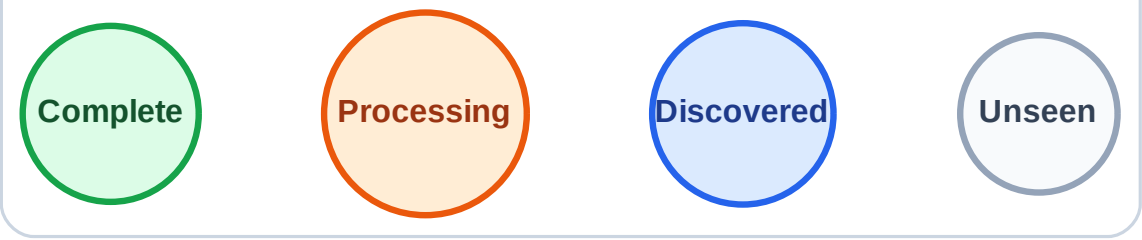
Visited: A, B, C, D, F, G, I, O



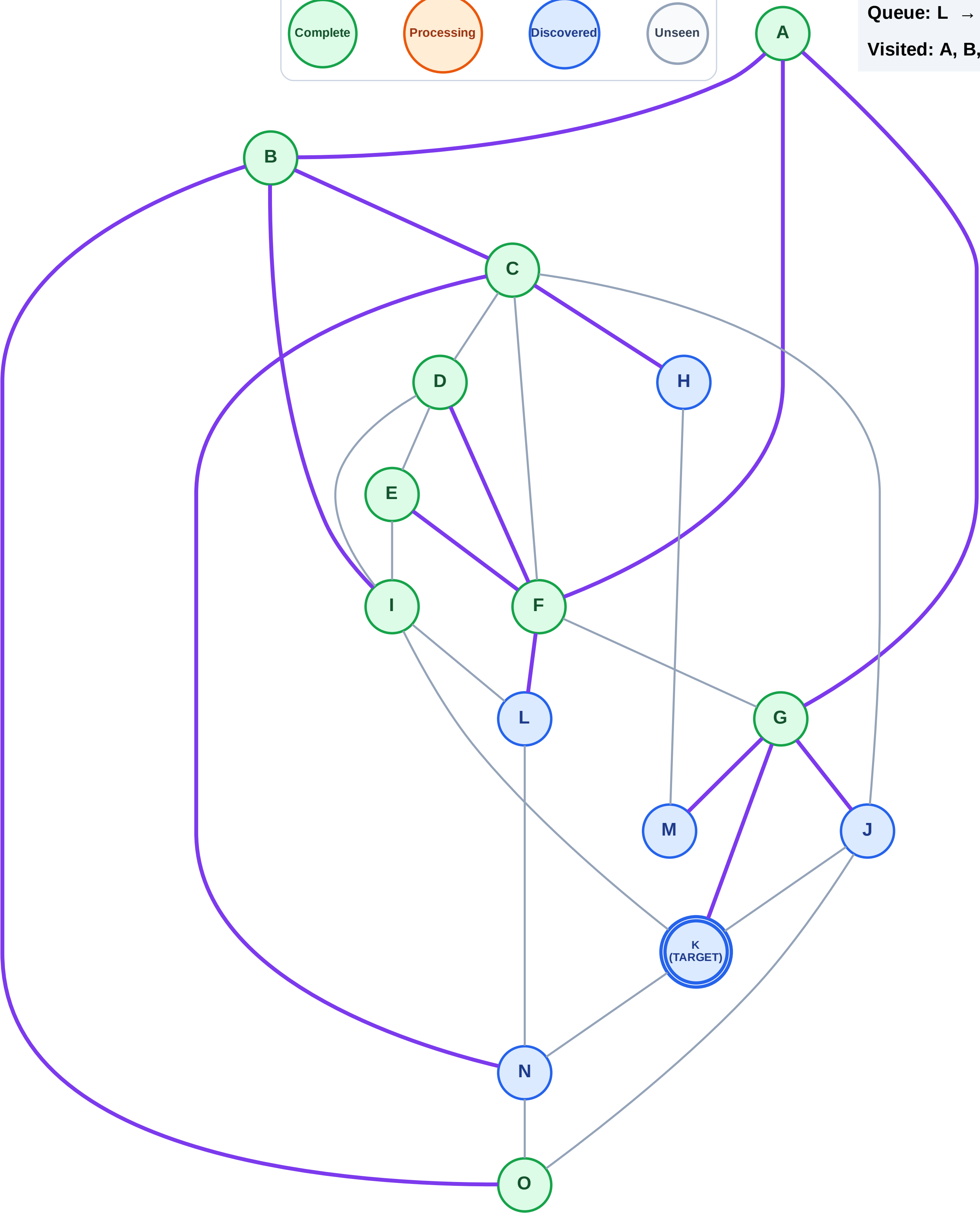
BFS Traversal

Step 19: No new neighbors from E

Legend



Queue: L → J → K → M → H → N
Visited: A, B, C, D, E, F, G, I, O



Step 20: Process node L

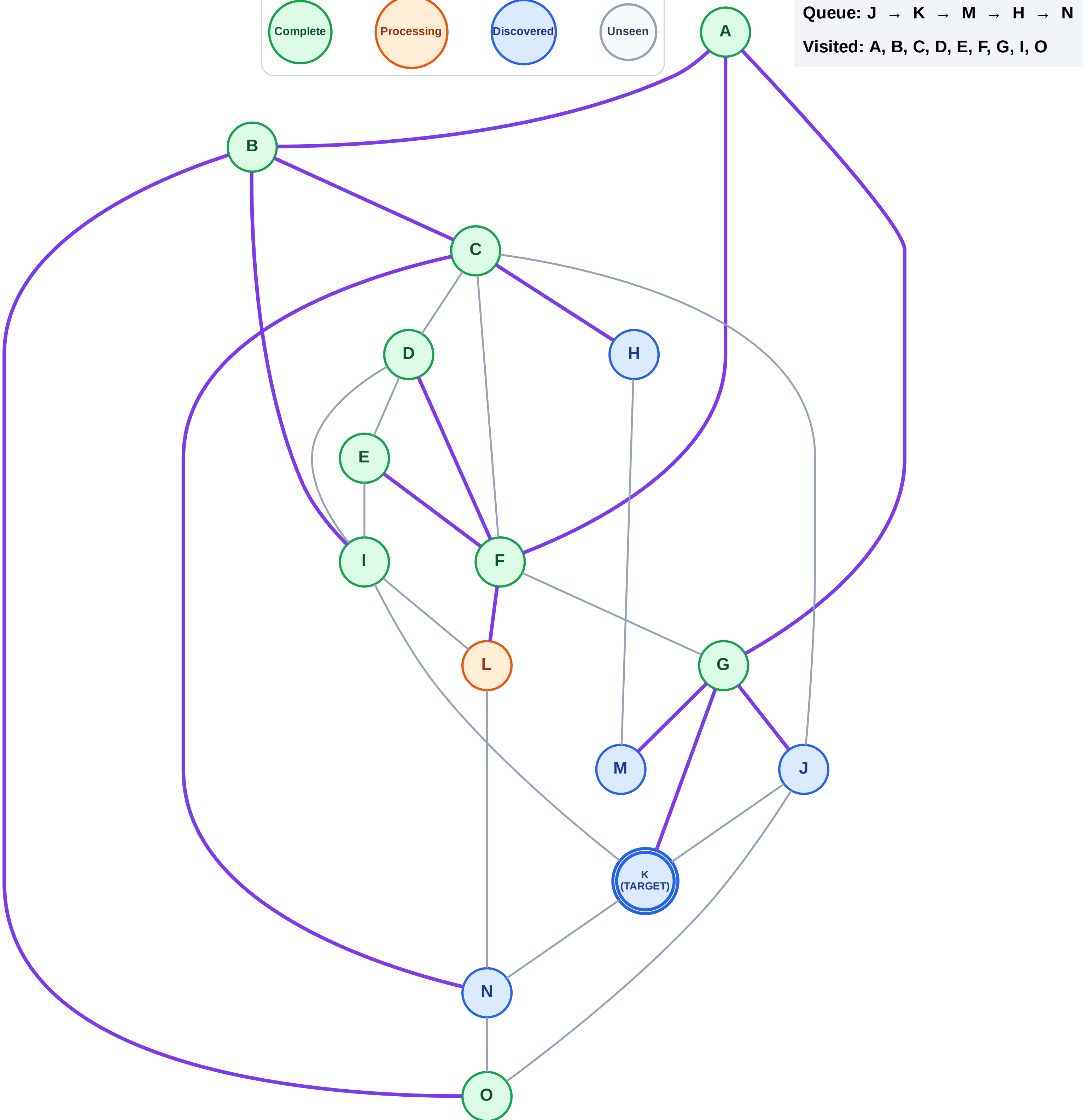
gend

Legend

Complete Processing Discovered Unseen

Unseen

Visited: A, B, C, D, E, F, G, I, O



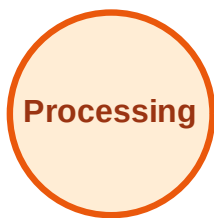
BFS Traversal

Step 21: No new neighbors from L

Legend



Complete



Processing

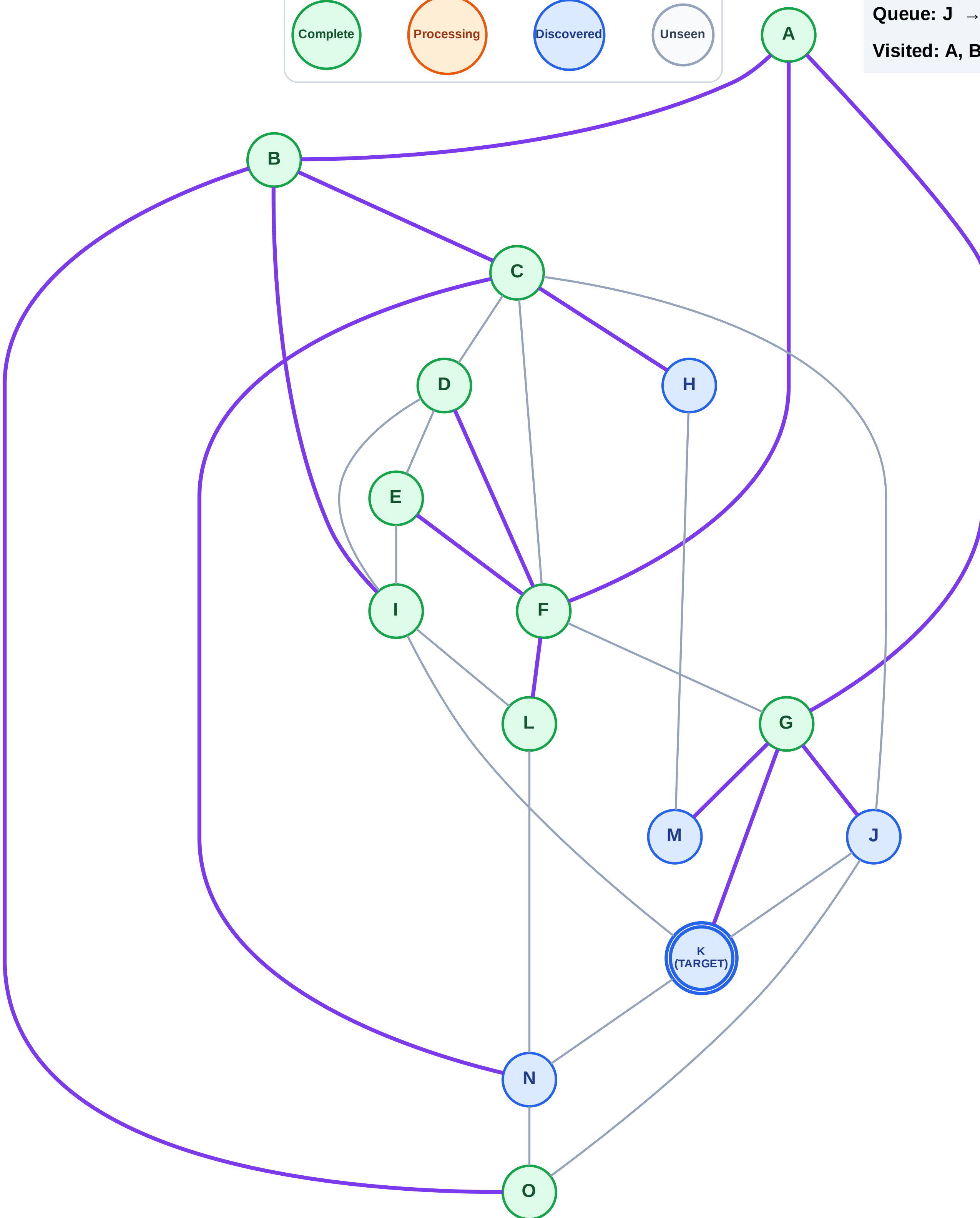


Discovered



Unseen

Queue: J → K → M → H → N
Visited: A, B, C, D, E, F, G, I, L, O



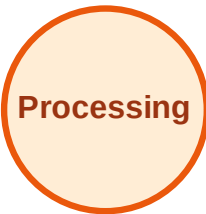
BFS Traversal

Step 22: Process node J

Legend



Complete



Processing



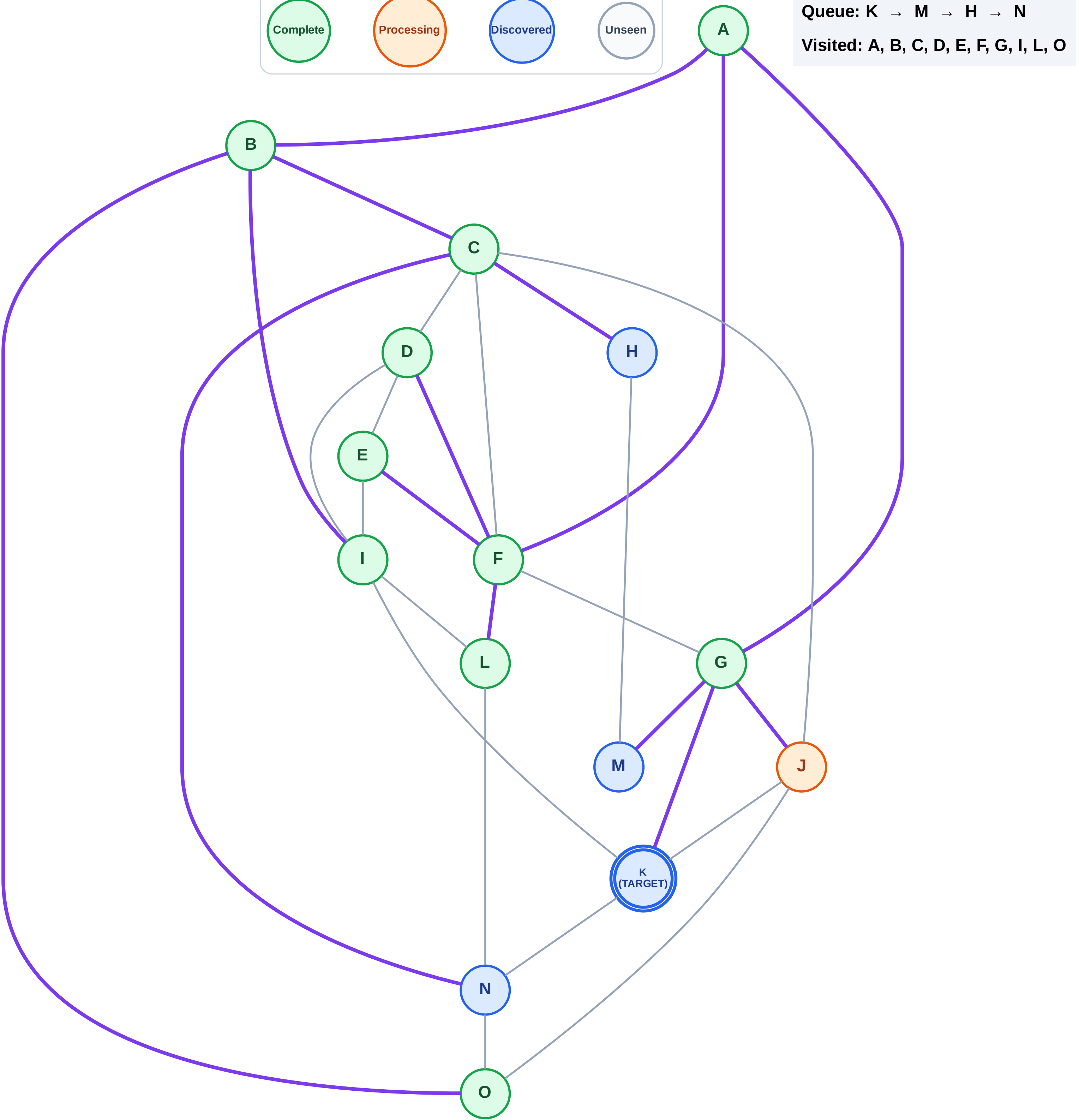
Discovered



Unseen

Queue: K → M → H → N

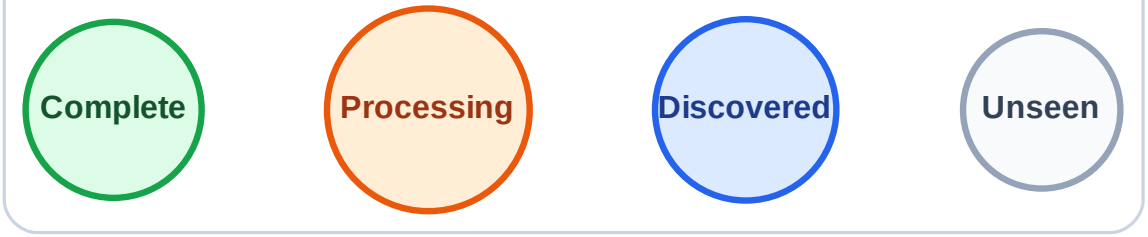
Visited: A, B, C, D, E, F, G, I, L, O



BFS Traversal

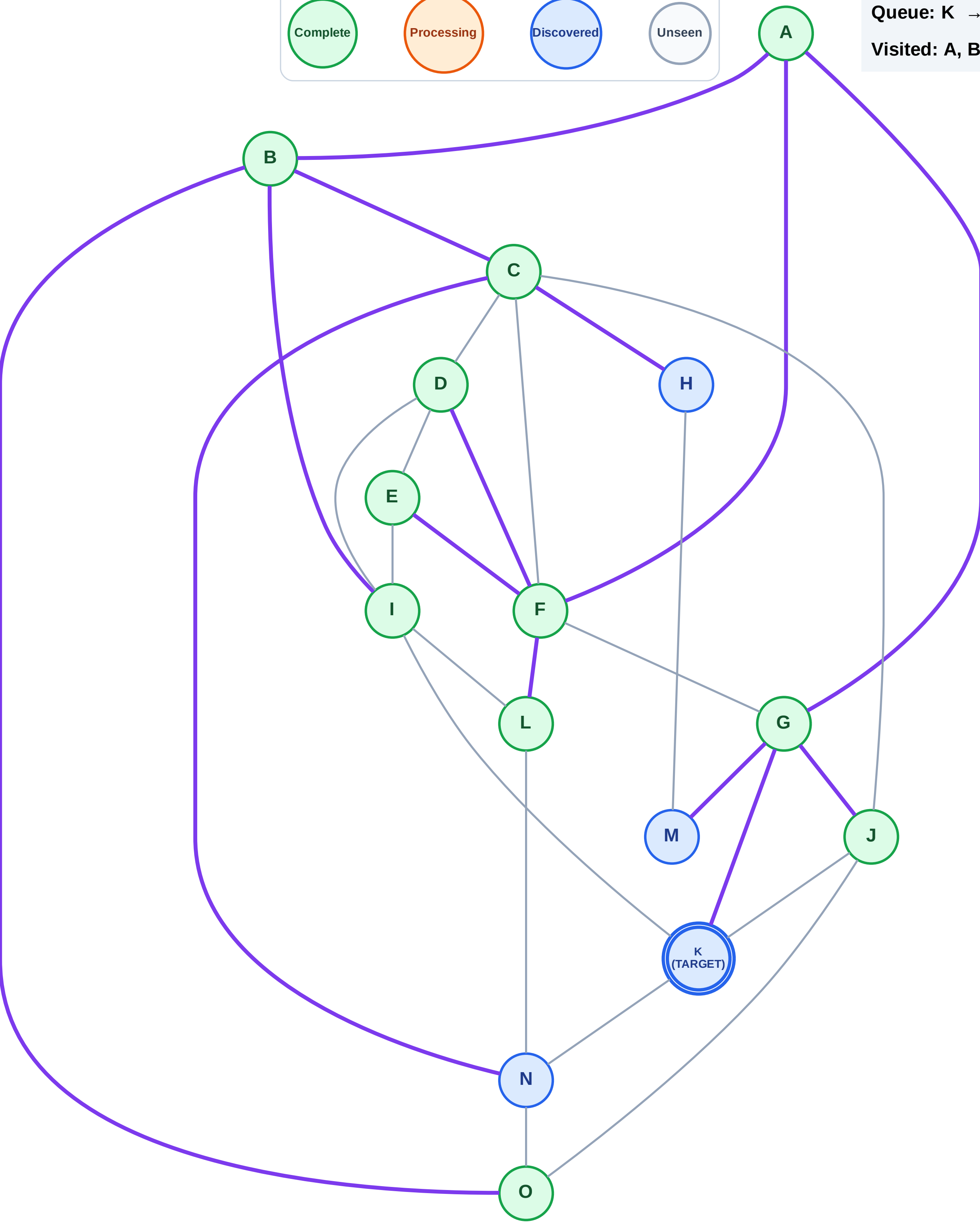
Step 23: No new neighbors from J

Legend



Queue: K → M → H → N

Visited: A, B, C, D, E, F, G, I, J, L, O



BFS Traversal

Step 24: Process node K

Legend



Complete



Processing



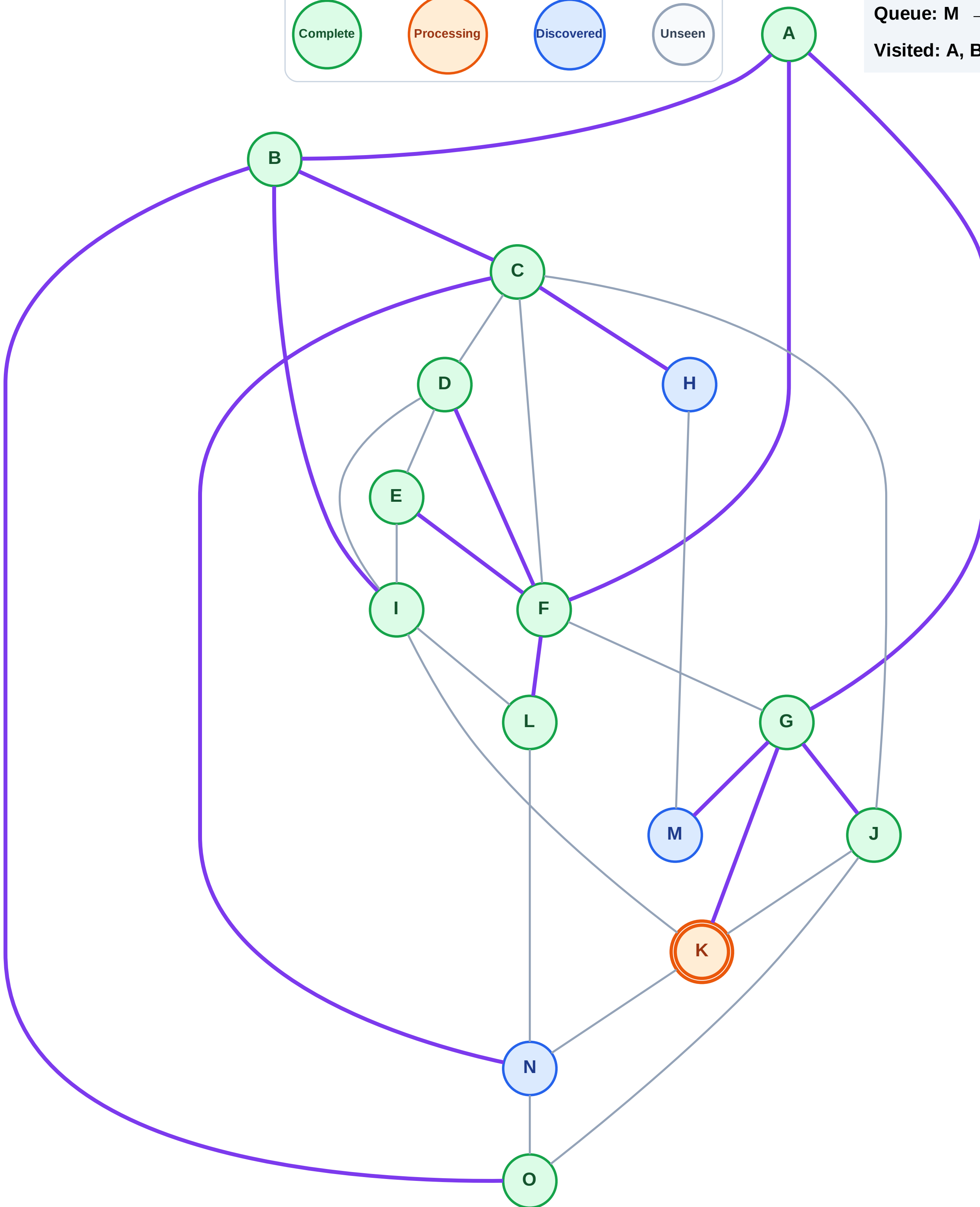
Discovered



Unseen

Queue: M → H → N

Visited: A, B, C, D, E, F, G, I, J, L, O



BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend

Complete

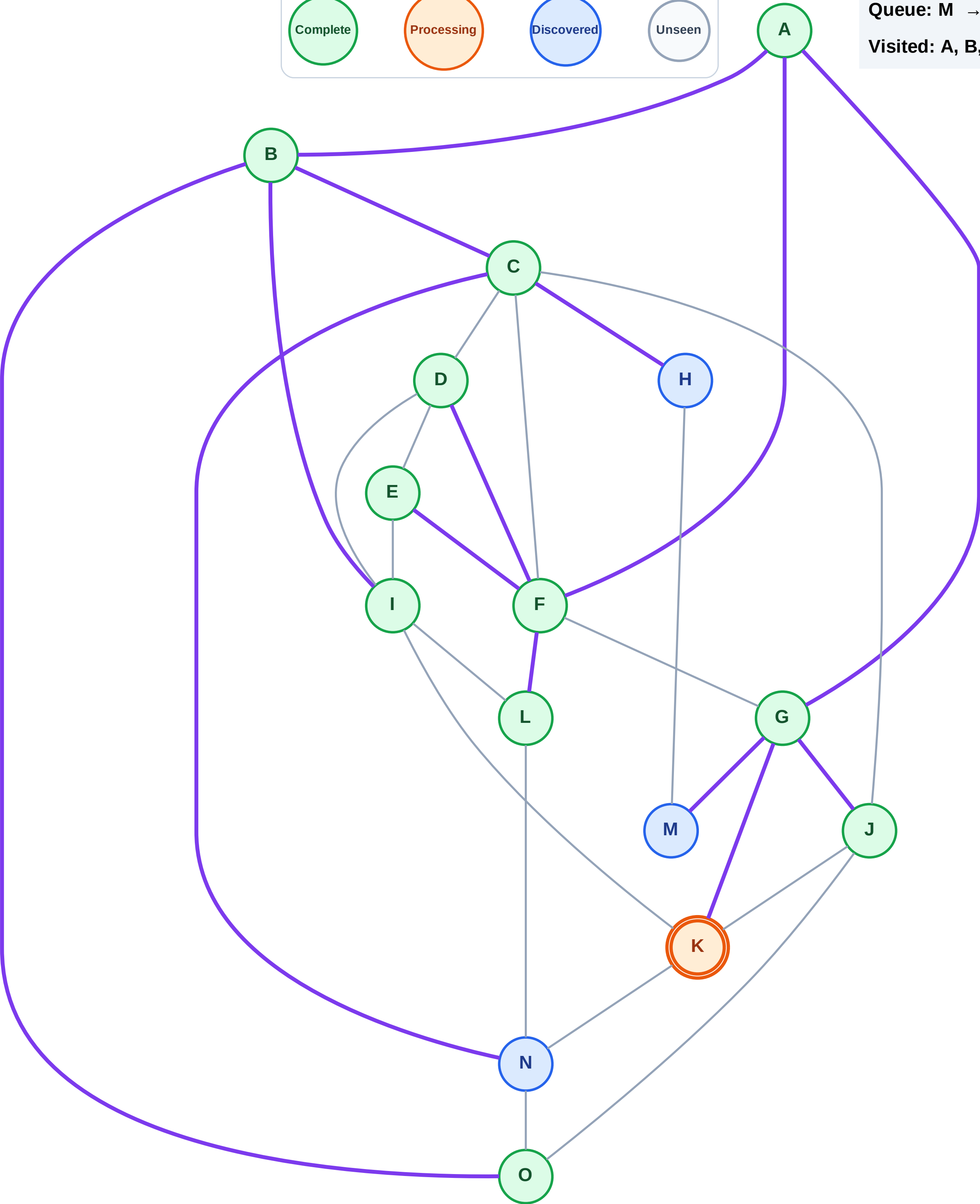
Processing

Discovered

Unseen

Queue: M → H → N

Visited: A, B, C, D, E, F, G, I, J, K, L, O



Step 1: Start at A (Seeking target node K)

Step 1: Start at A (Seeking target node K)

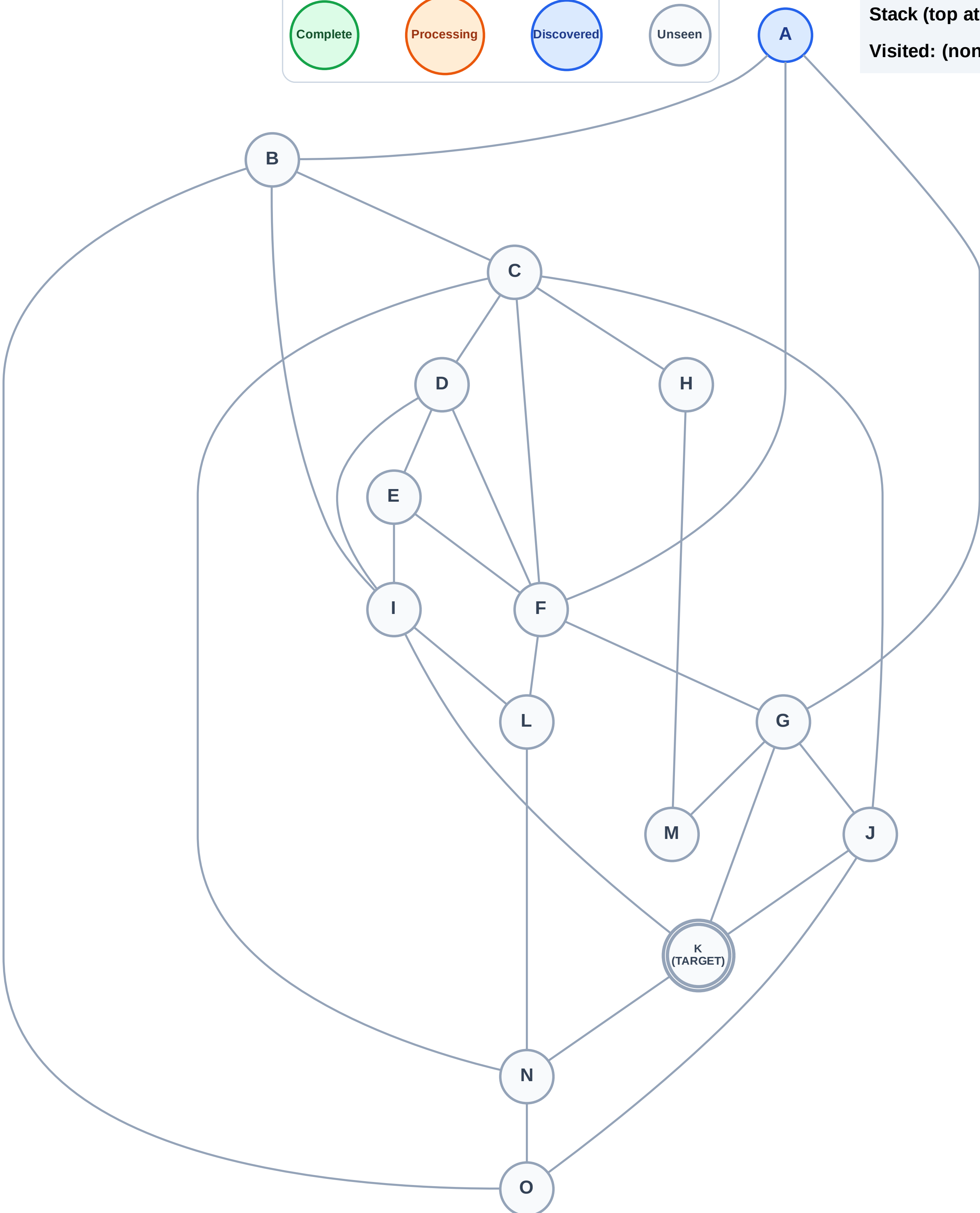
Legend

Complete Processing Discovered Unseen

Processing

Unseen

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

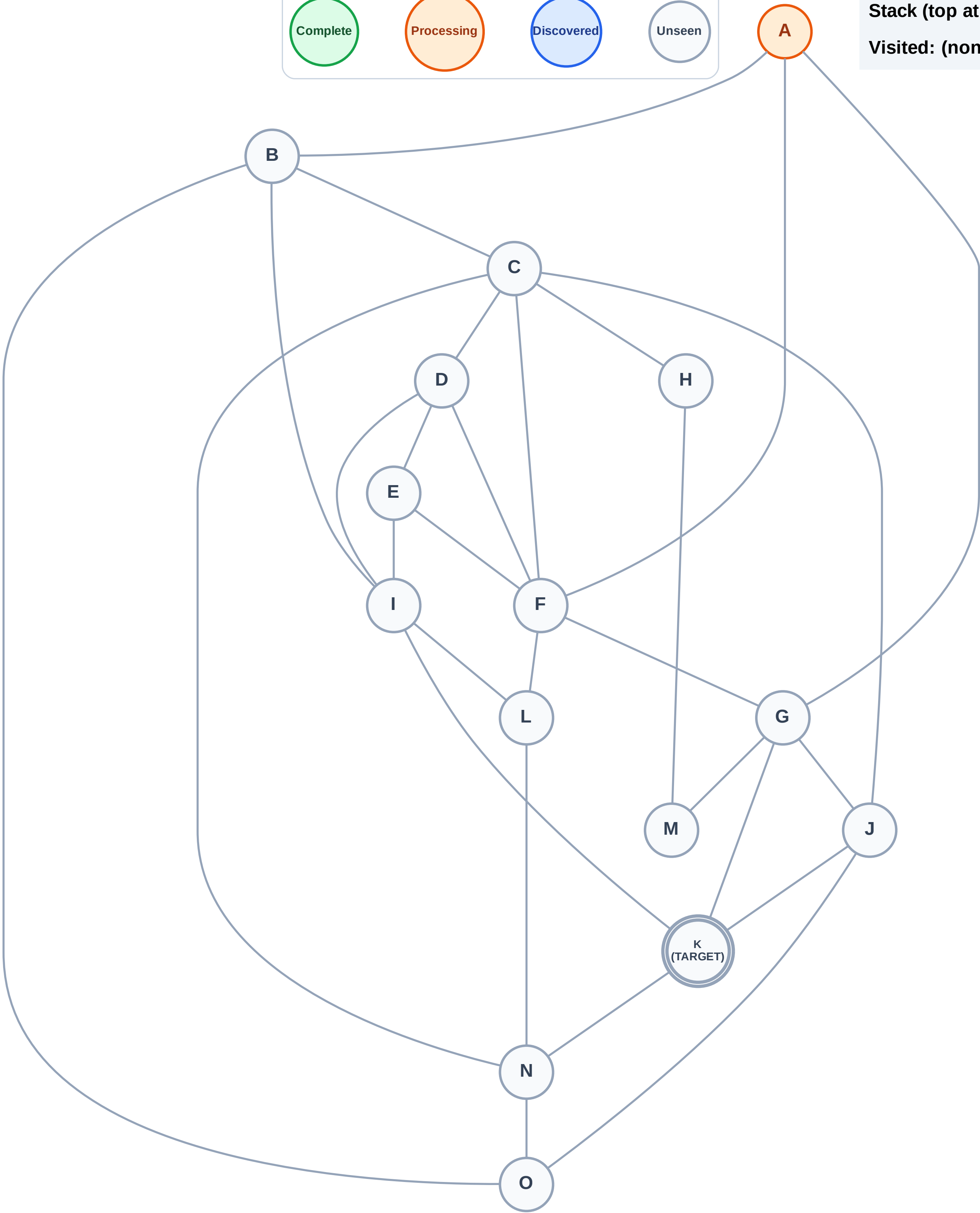
Processing

Discovered

Unseen

Stack (top at right): (empty)

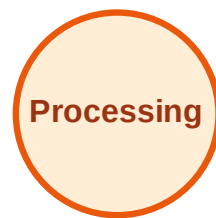
Visited: (none)



DFS Traversal

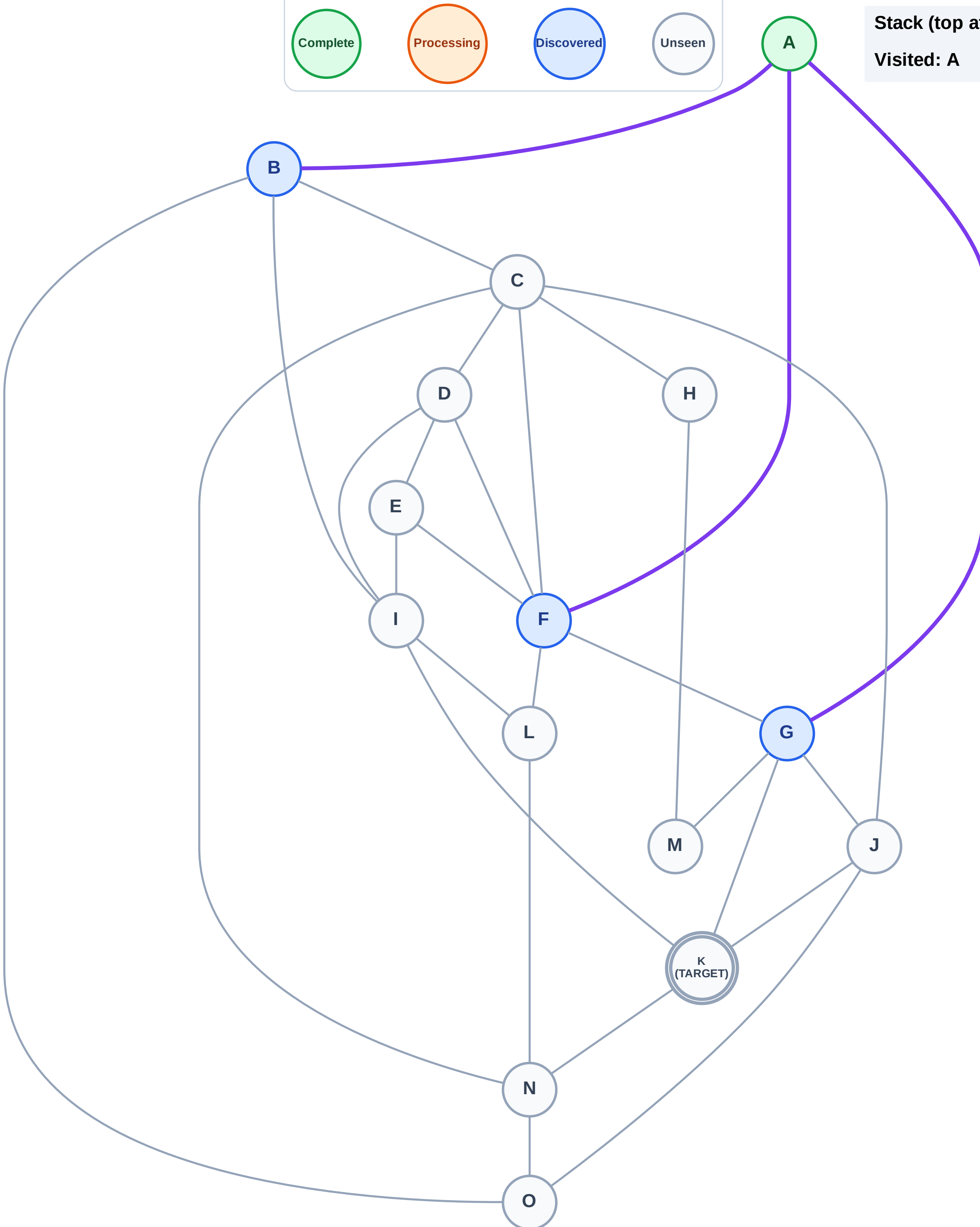
Step 3: Discover G, F, B from A

Legend



Stack (top at right): G | F | B

Visited: A

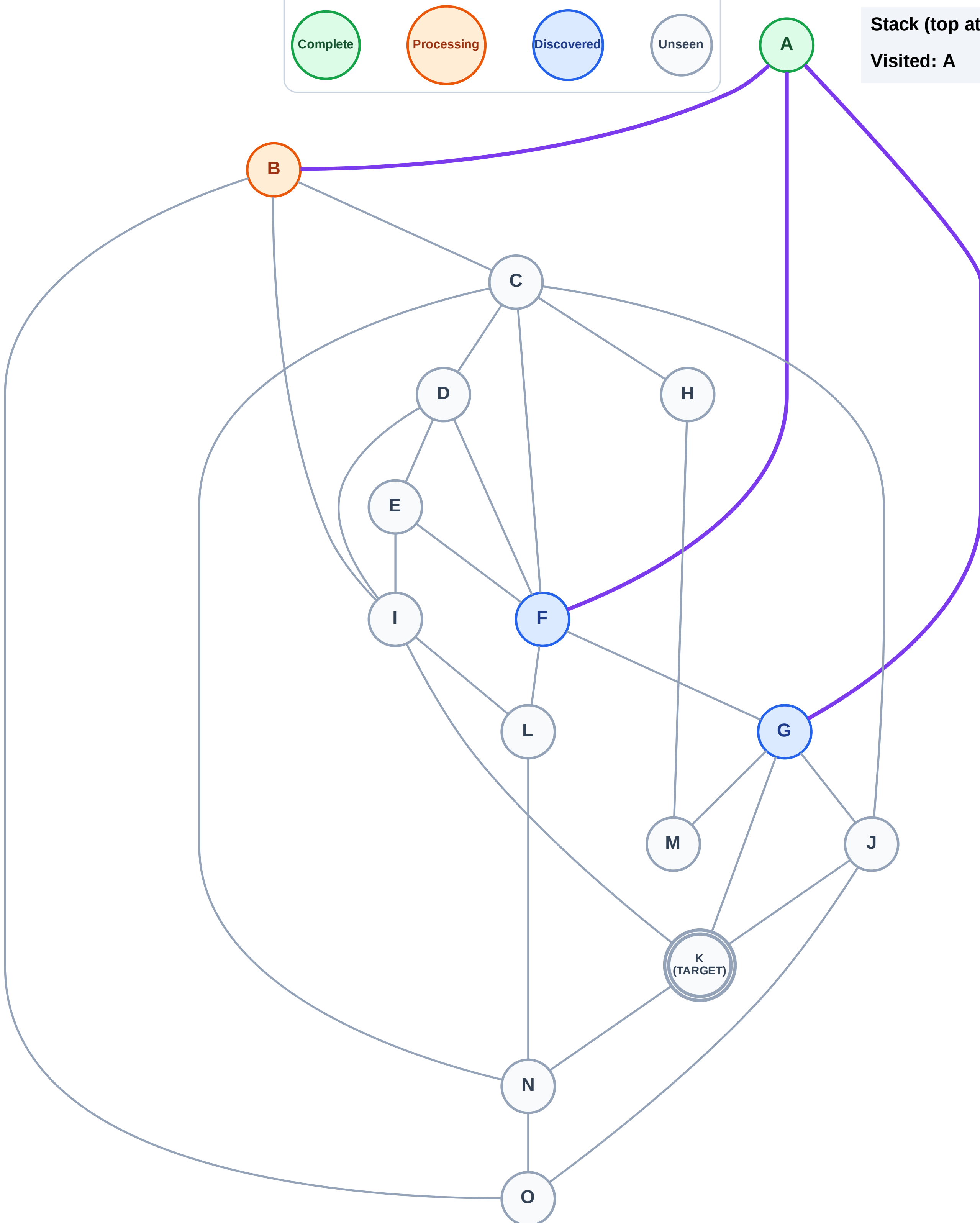


Step 4: Process node B

Legend



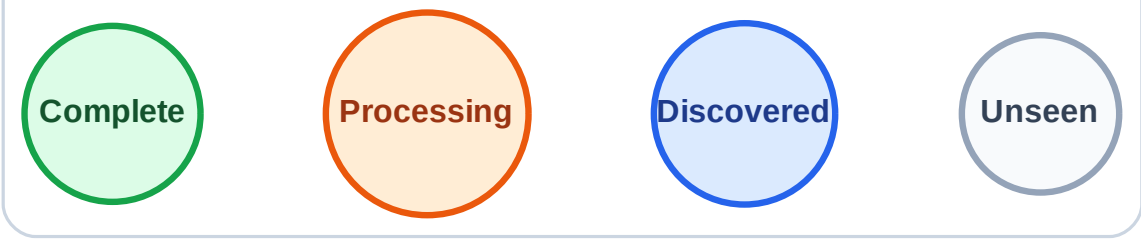
Visited: A



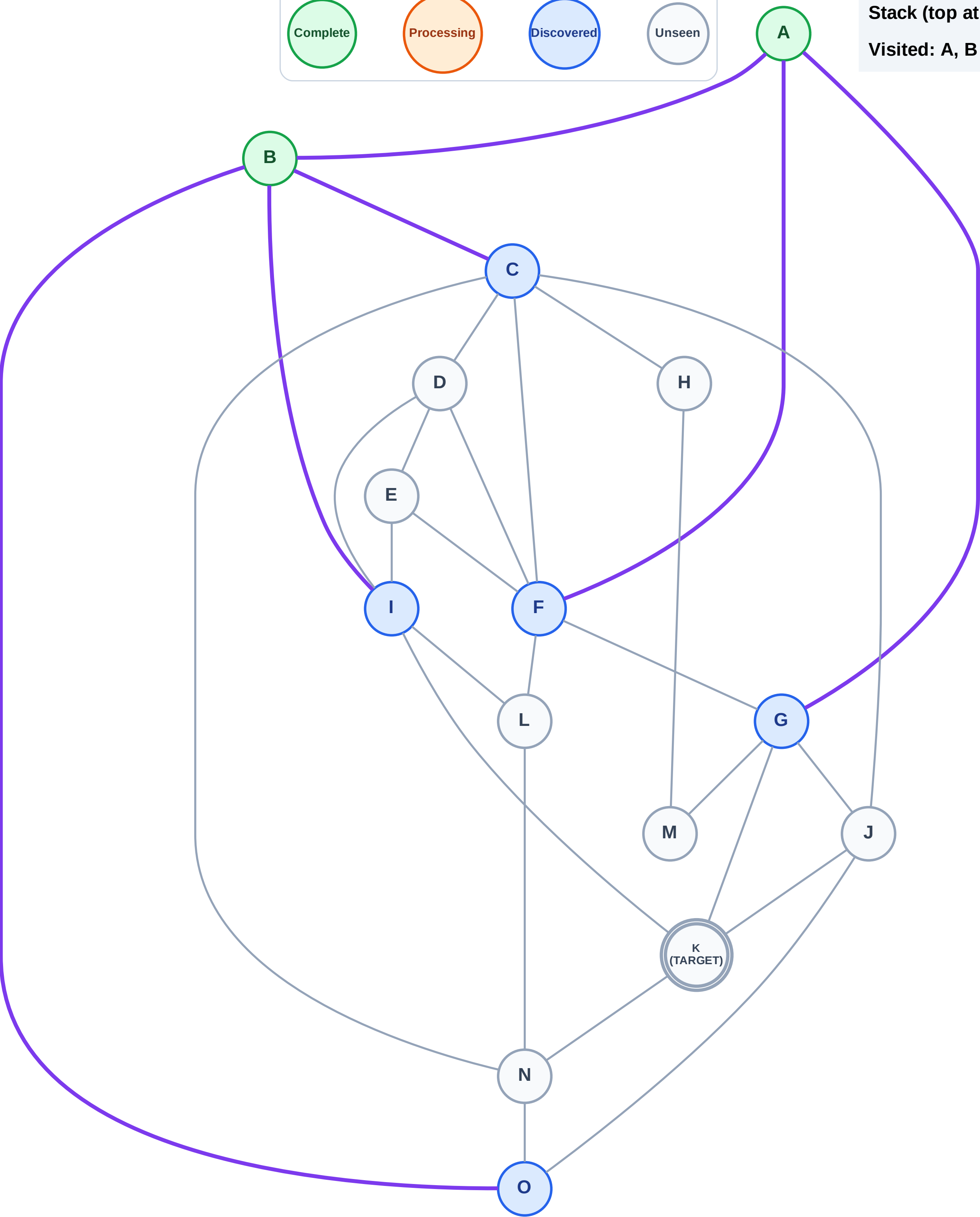
DFS Traversal

Step 5: Discover O, I, C from B

Legend



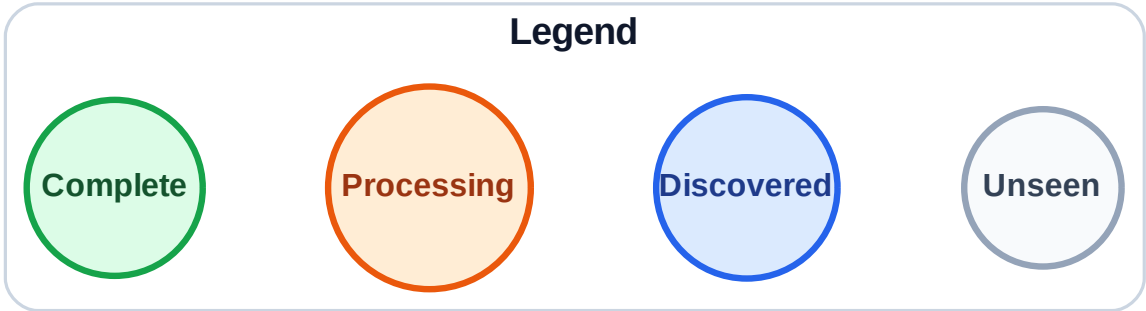
Stack (top at right): G | F | O | I | C
Visited: A, B



DFS Traversal

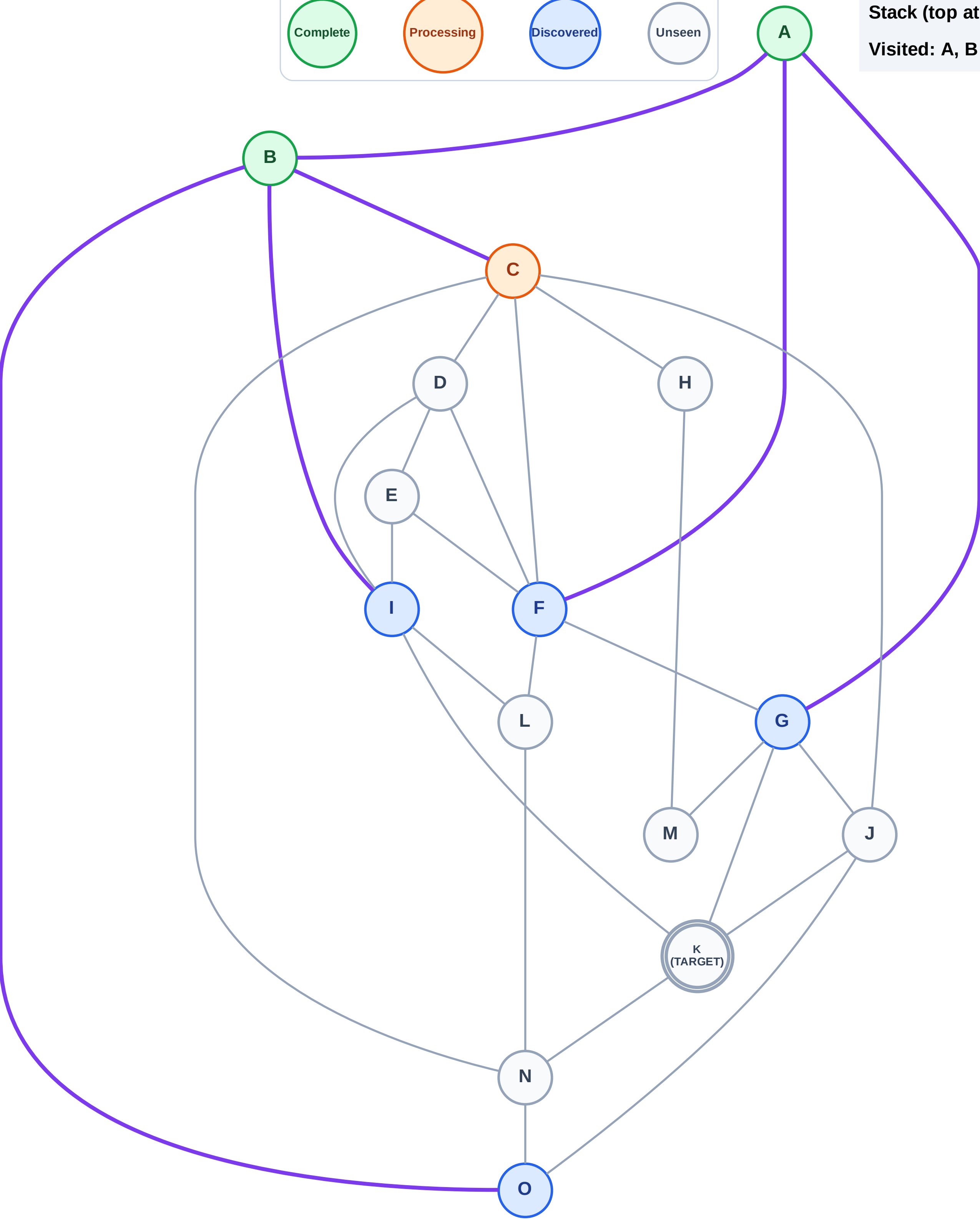
Step 6: Process node C

Legend



Stack (top at right): G | F | O | I

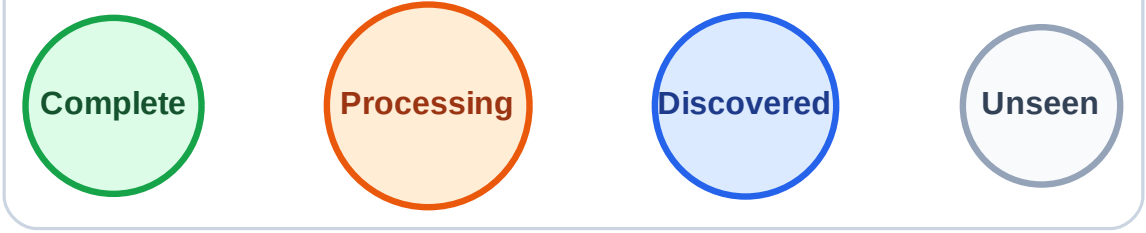
Visited: A, B



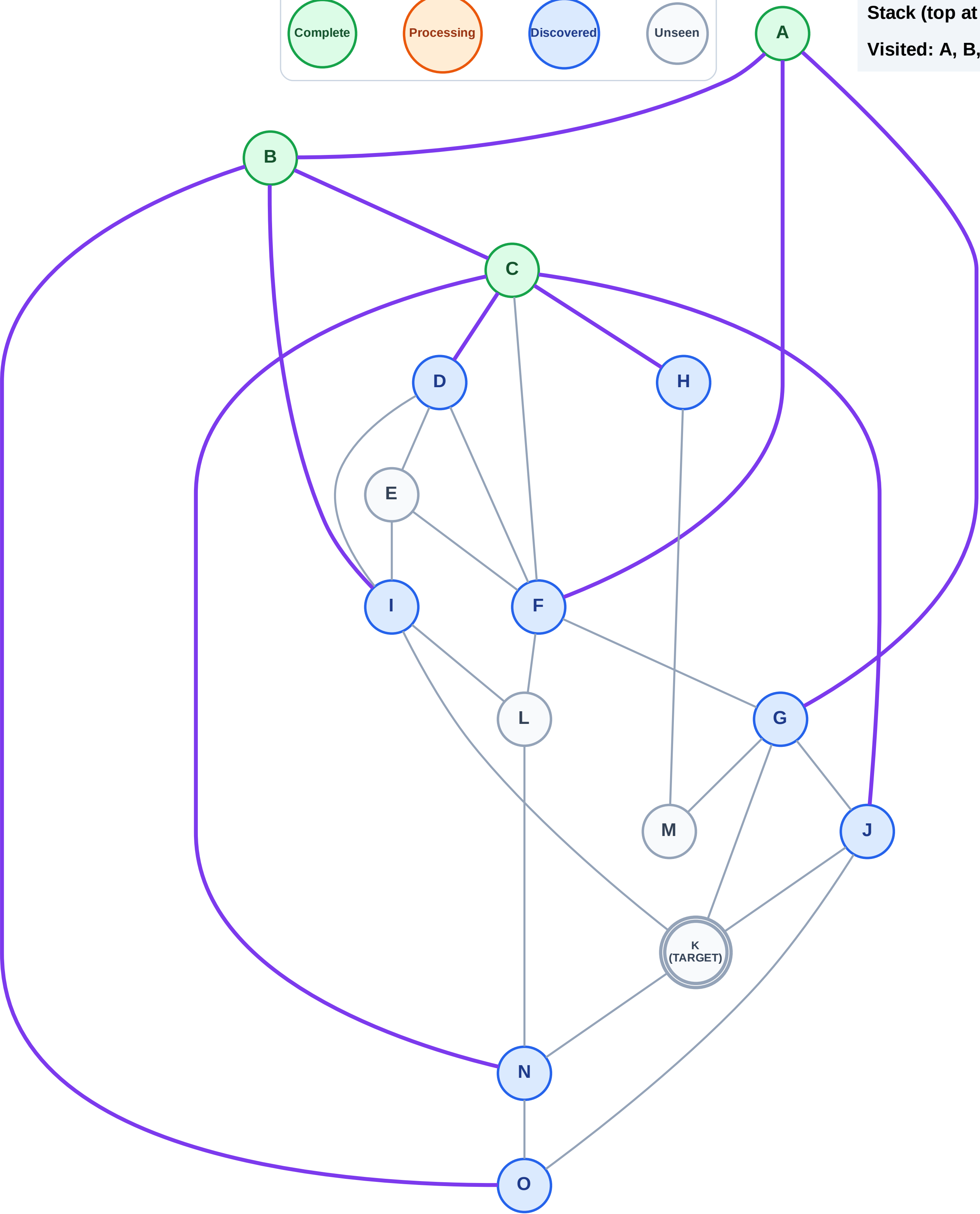
DFS Traversal

Step 7: Discover N, J, H, D from C

Legend



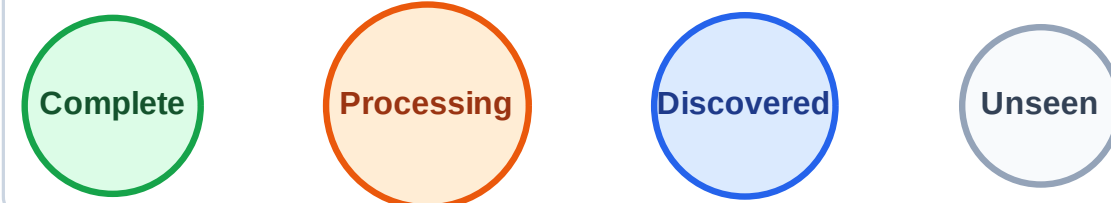
Stack (top at right): G | F | O | I | N | J | H | D
Visited: A, B, C



Step 8: Process node D

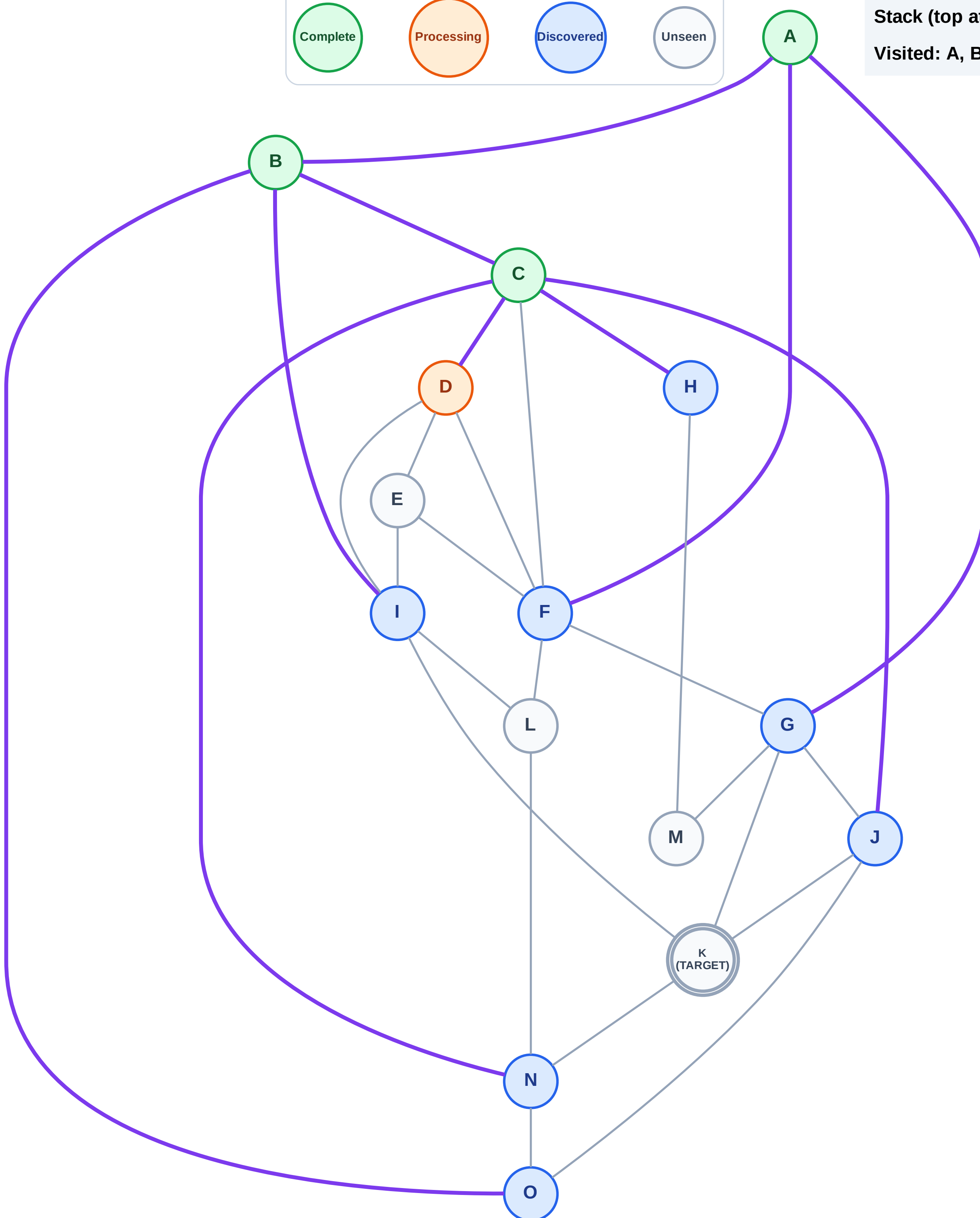
Page 10 of 10

Legend



Stack (top at right): G | F | O | I | N | J | H

Visited: A, B, C



DFS Traversal

Step 9: Discover E from D

Legend

Complete

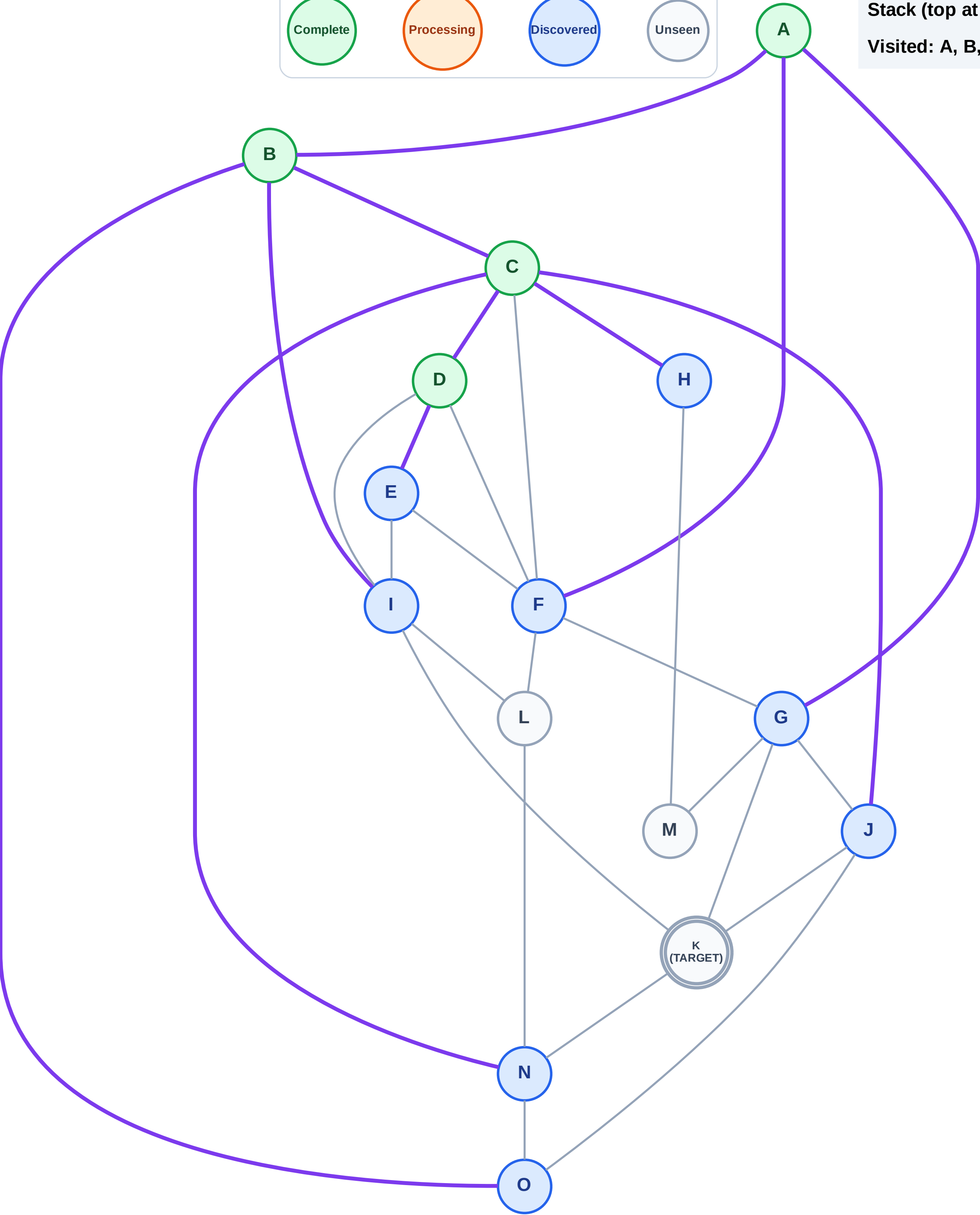
Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N | J | H | E

Visited: A, B, C, D



DFS Traversal

Step 10: Process node E

Legend

Complete

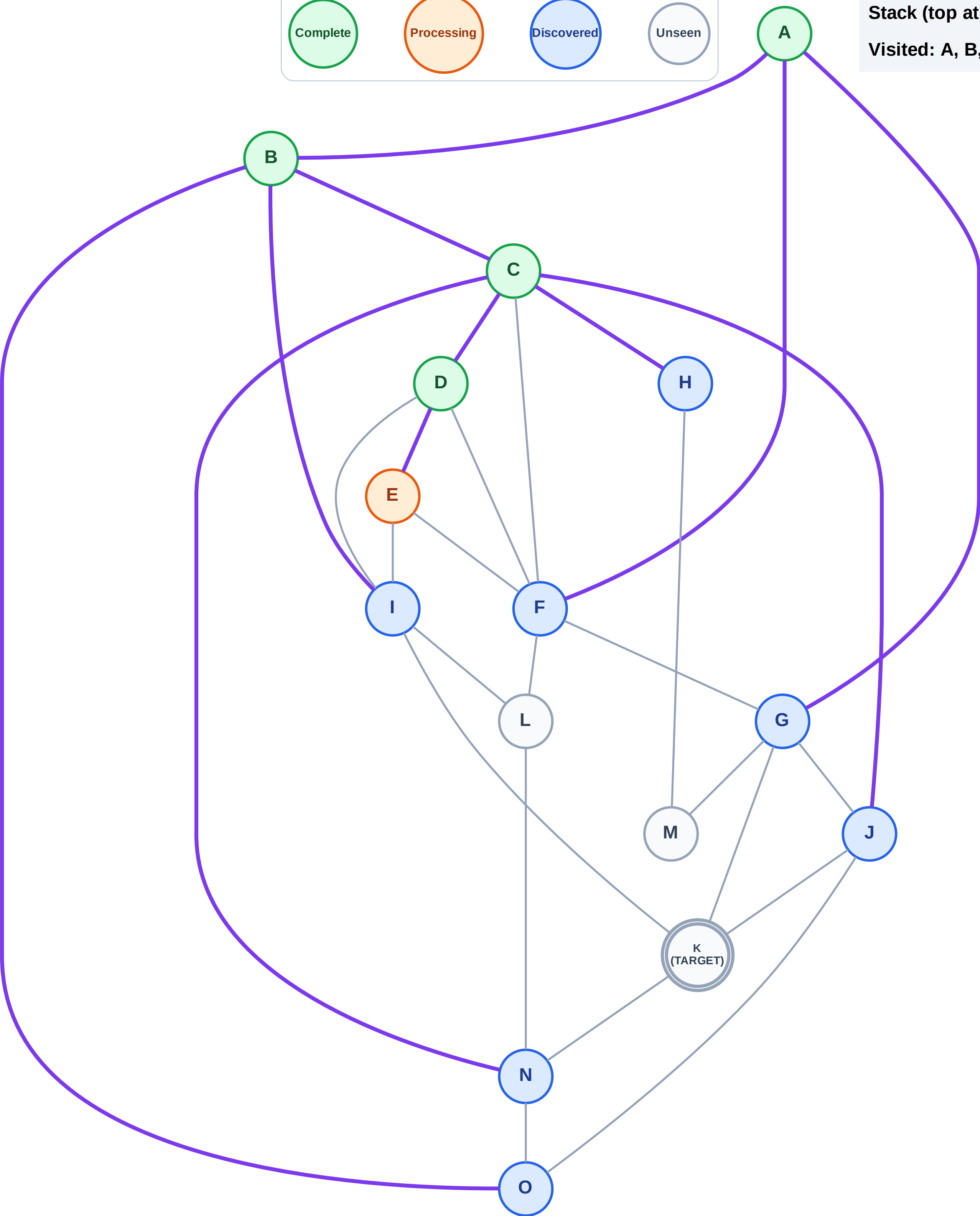
Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N | J | H

Visited: A, B, C, D



DFS Traversal

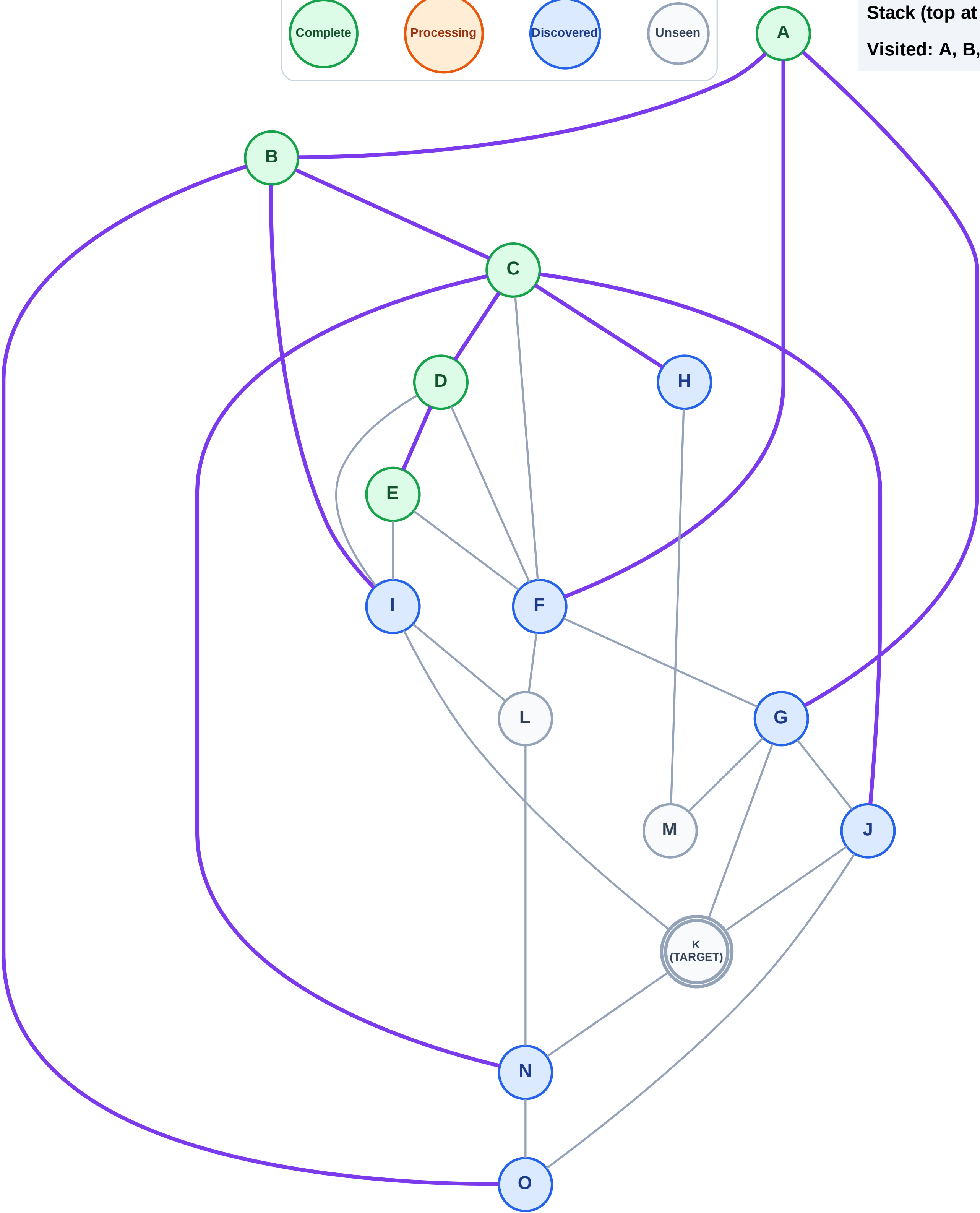
Step 11: No new neighbors from E

Legend



Stack (top at right): G | F | O | I | N | J | H

Visited: A, B, C, D, E



Step 12: Process node H

Copyright © 2016 Pearson Education, Inc. or its affiliate(s). All rights reserved.

Legend

Complete

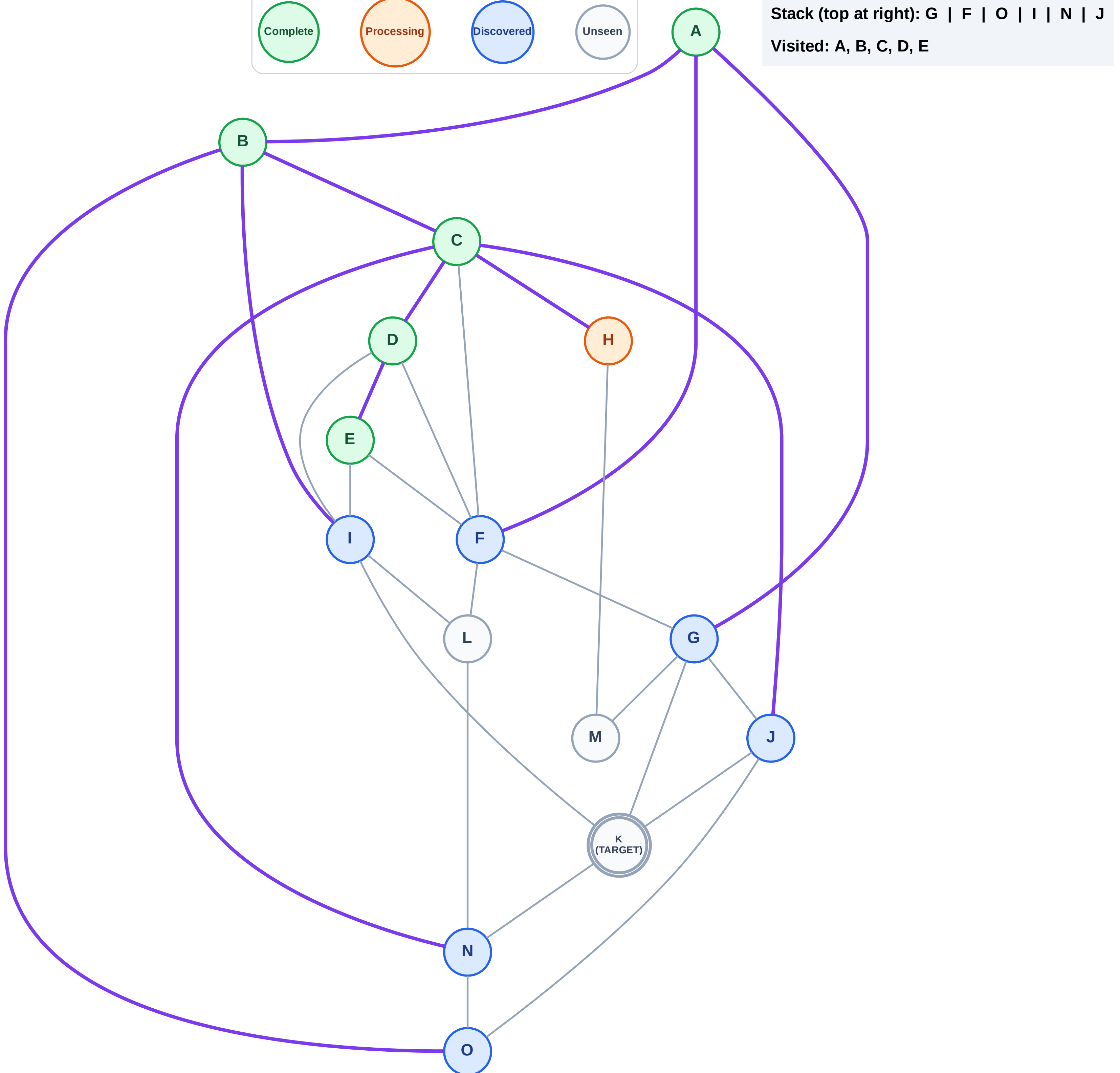
Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N | J

Visited: A, B, C, D, E



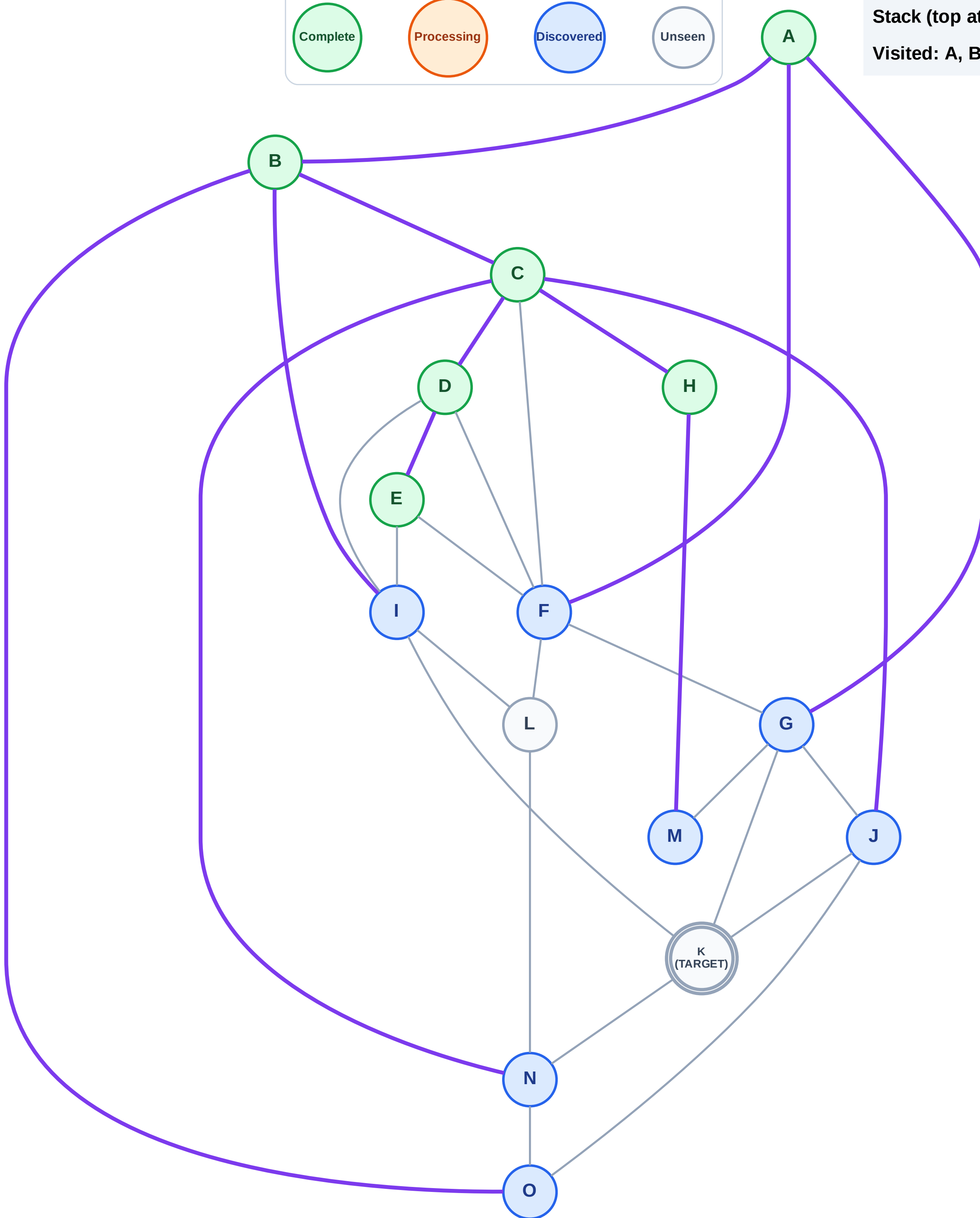
DFS Traversal

Step 13: Discover M from H

Legend



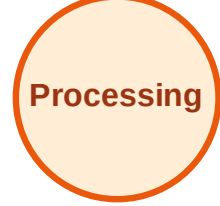
Stack (top at right): G | F | O | I | N | J | M
Visited: A, B, C, D, E, H



DFS Traversal

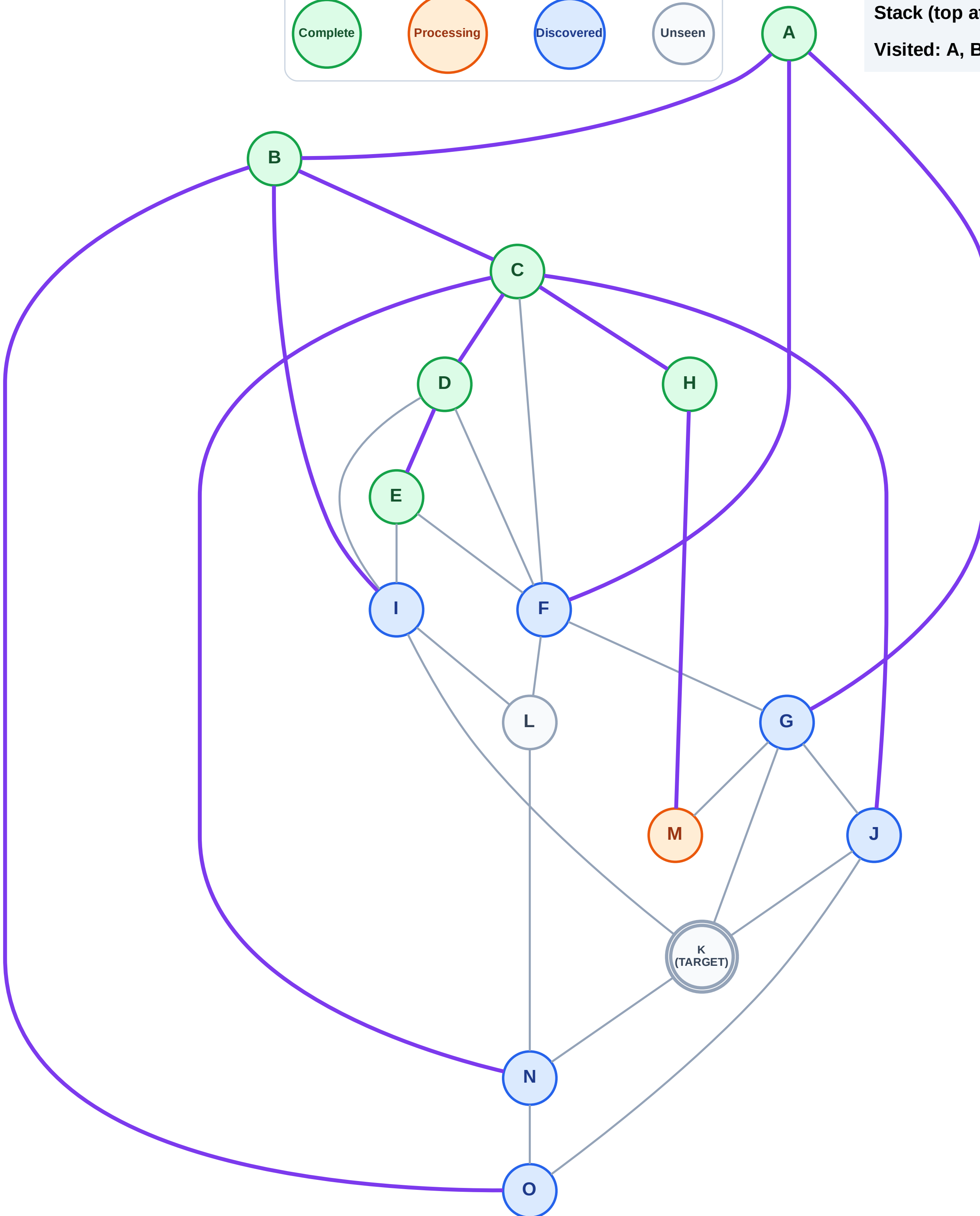
Step 14: Process node M

Legend



Stack (top at right): G | F | O | I | N | J

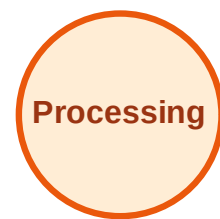
Visited: A, B, C, D, E, H



DFS Traversal

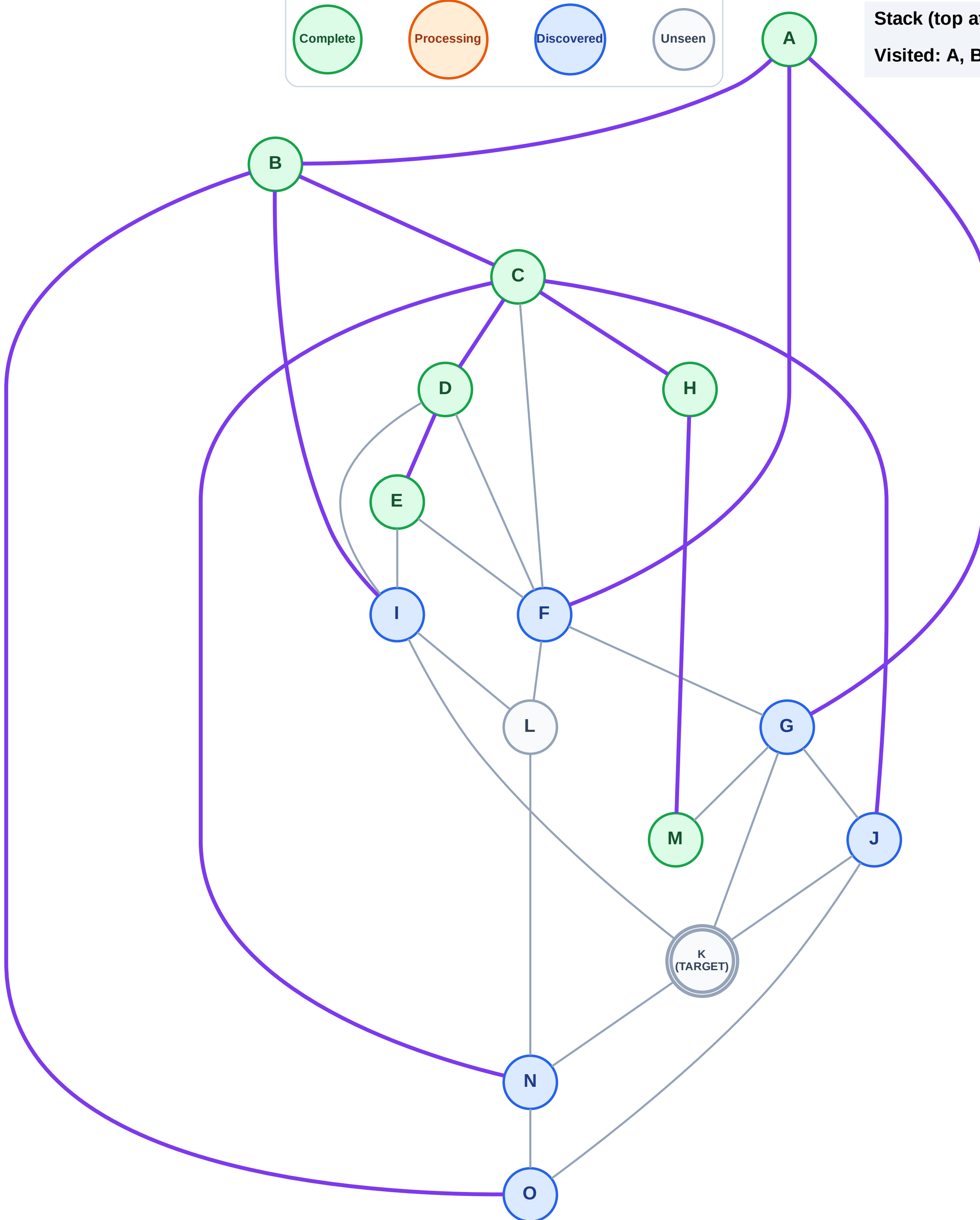
Step 15: No new neighbors from M

Legend



Stack (top at right): G | F | O | I | N | J

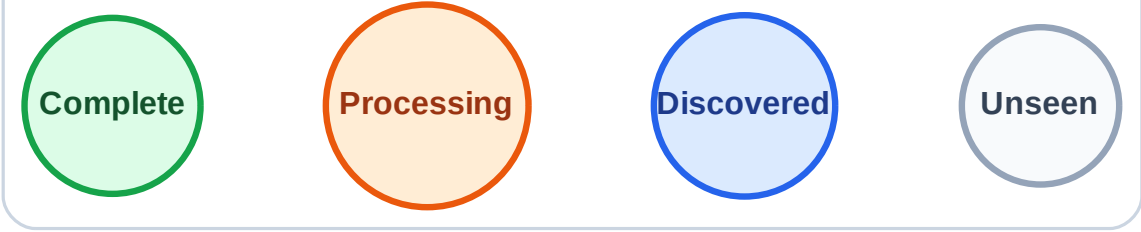
Visited: A, B, C, D, E, H, M



DFS Traversal

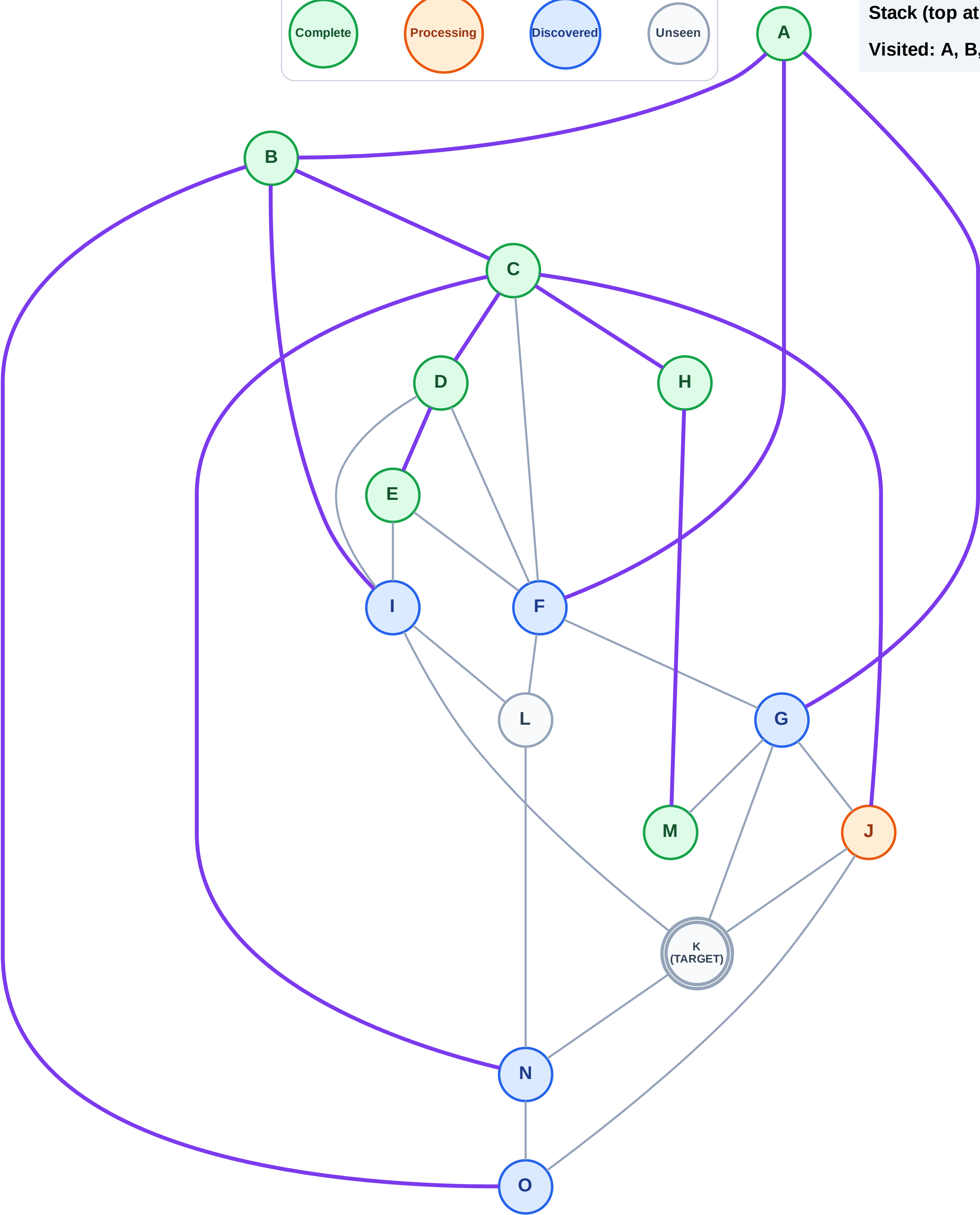
Step 16: Process node J

Legend



Stack (top at right): G | F | O | I | N

Visited: A, B, C, D, E, H, M

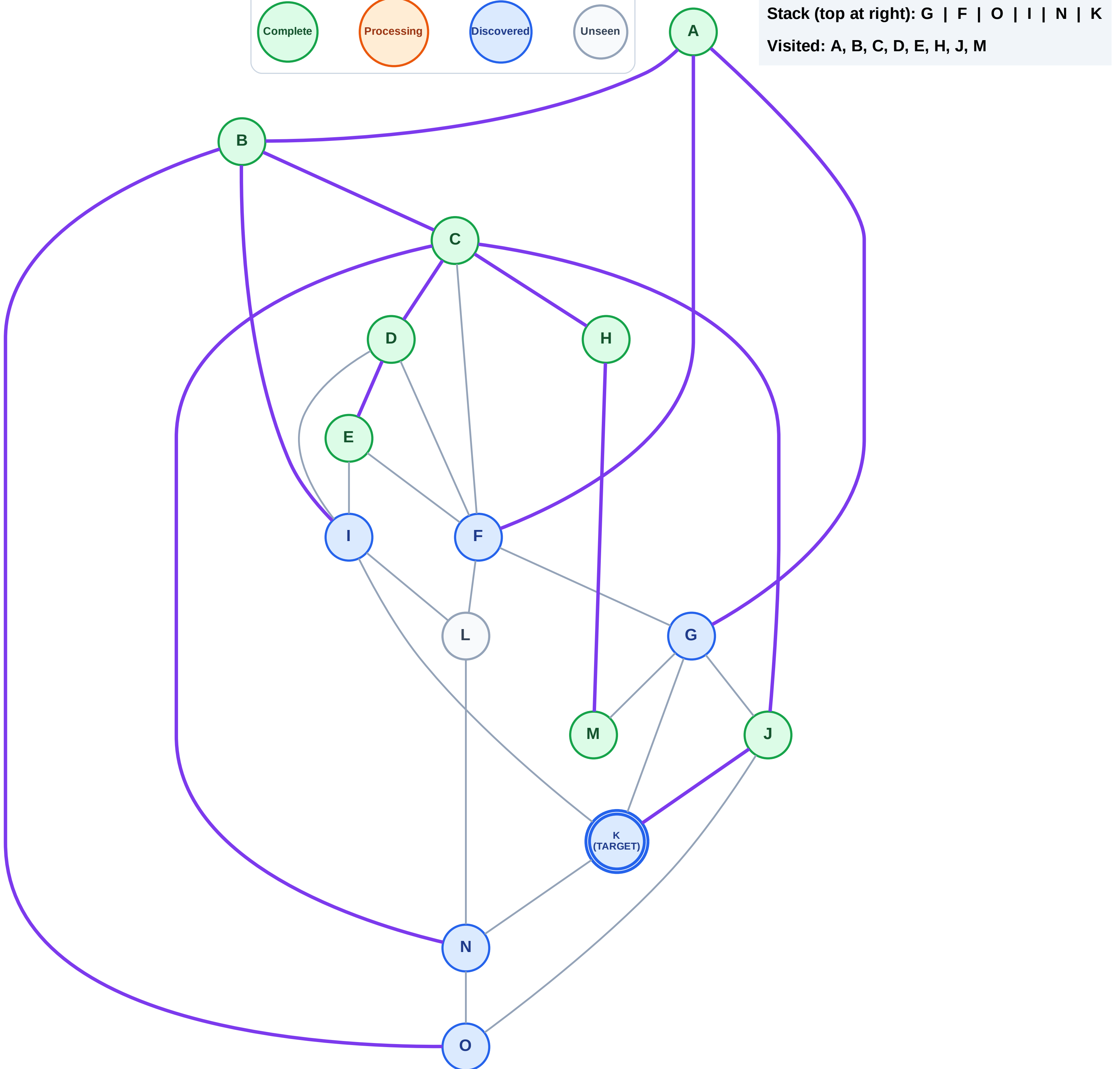


Step 17: Discover K from J

Legend

Unseen

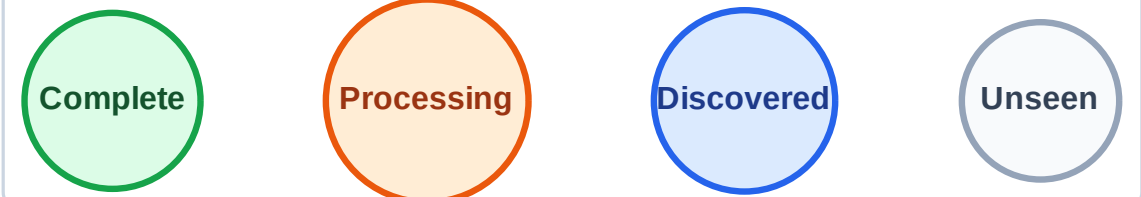
Visited: A, B, C, D, E, H, J, M



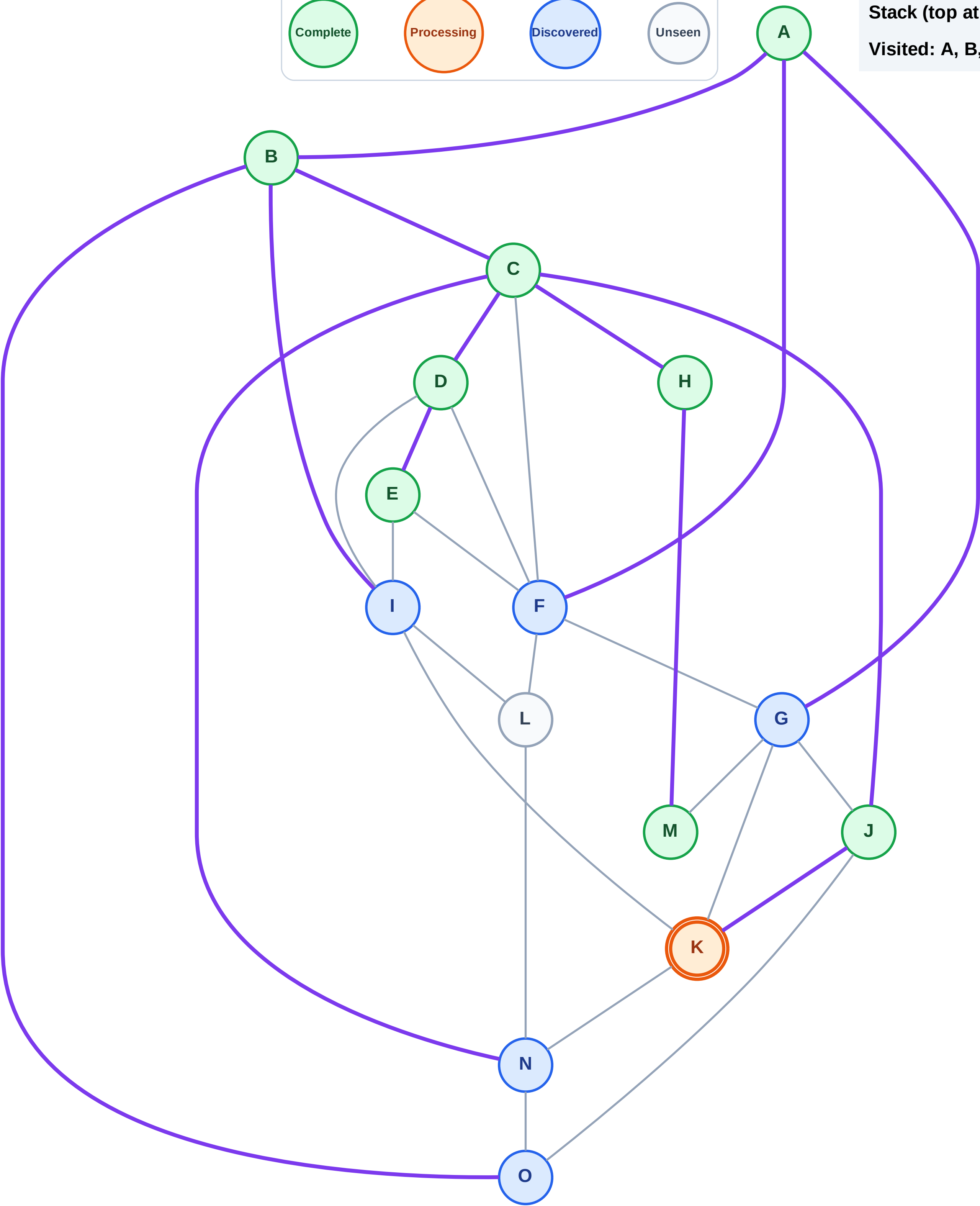
DFS Traversal

Step 18: Process node K

Legend



Stack (top at right): G | F | O | I | N
Visited: A, B, C, D, E, H, J, M



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (4 hops): A → B → C → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N
Visited: A, B, C, D, E, H, J, K, M

